

Lecture Notes on

Finite Element Methods in

Fluid Mechanics

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Chapter 1

The finite element method for Poisson's problem

This chapter is mainly intended to introduce the notation and the basic concepts to be used in the rest of the course. The material is quite condensed. It is highly desirable that the reader has some background in the finite element method, both in the formulation and in the implementation. Likewise, even if the most common functional spaces appearing in the finite element theory are introduced, it is very convenient to consult other sources to get familiar with them.

{chap1}

1.1 Problem statement

{sec:c1-pr}

1.1.1 Boundary value problem

Let $\Omega \subset \mathbb{R}^d$, $d = 1, 2, 3$, be the domain where the problem is posed. Unless otherwise specified, we will consider this domain bounded and regular enough, so that the operations we wish to perform on it are well defined. In particular, we will assume that the boundary $\Gamma = \partial\Omega$ is locally Lipschitz, with external unit normal $\mathbf{n}$ well defined almost everywhere (i.e., except in a set of zero measure).

The problem we consider in this chapter is the following: find a scalar function $u : \Omega \rightarrow \mathbb{R}$ such that

$$-\kappa\Delta u + \sigma u = f \quad \text{in } \Omega, \quad (1.1) \quad \{\text{eq:c1eq}\}$$

$$u = \bar{u} \quad \text{on } \Gamma_u, \quad (1.2) \quad \{\text{eq:c1cu}\}$$

$$\kappa \frac{\partial u}{\partial \mathbf{n}} + \beta u = g \quad \text{on } \Gamma_g. \quad (1.3) \quad \{\text{eq:c1cg}\}$$

The notation used here and in the rest of these notes is as follows. We consider $\Gamma = \overline{\Gamma_u} \cup \overline{\Gamma_g}$ (with $\Gamma_u \cap \Gamma_g = \emptyset$), where Γ_u is the part of Γ where u itself is prescribed to a given function $\bar{u}$, and Γ_g is the part of Γ where the flux of u , given by $\kappa \frac{\partial u}{\partial \mathbf{n}} = \kappa \mathbf{n} \cdot \nabla u$, is prescribed to a given function g minus a term proportional to u . Functions $\bar{u}$, g and f are the *data* of the problem.

For simplicity, we will consider the physical properties κ , σ and β constant, although the extension to bounded functions is easy. In the problems of interest, $\kappa > 0$, and we will require $\sigma \geq 0$. The case $\sigma < 0$ will not be considered here, as it requires a different treatment; the reason is that $-\sigma > 0$ could be an eigenvalue of $-\Delta$. Likewise, well-posedness of the problem holds if $\beta \geq 0$.

Let us discuss the boundary conditions in Eq. (1.2) and Eq. (1.3). The former is called of **Dirichlet** type, whereas the latter is called of **Robin** type. In fact, the most common case is $\beta = 0$, when the boundary condition is called of **Neumann** type. If $\beta = 0$, $\sigma = 0$ and $\Gamma_u = \emptyset$, integrating Eq. (1.1) over

Ω and using Eq. (1.3) it is immediately checked that the data f and g need to satisfy the compatibility condition:

$$\int_{\Omega} f = - \int_{\Omega} \kappa \Delta u = - \int_{\partial\Omega} \kappa \frac{\partial u}{\partial n} = - \int_{\partial\Omega} g.$$

If this condition holds, the solution is unique up to constants. In the rest of cases considered, the solution exists and is unique, for appropriate regularity of the data. This solution is called **classical** if u is twice differentiable with continuity, at least piecewise, in the interior of Ω and differentiable up to the boundary (if $\Gamma_g = \emptyset$ continuity is enough), that is to say, $u \in \mathcal{C}_s^2(\Omega) \cap \mathcal{C}^1(\bar{\Omega})$.

1.1.2 A disclaimer about units

In these notes we will pay attention to units. All the results to be presented will be invariant under changes of units. This is not important in general in mathematical analysis, where usually one only cares about limits and asymptotic behaviour of functions or sequences, but in *numerical* analysis it is crucial, as numerical values do matter. Depending on the machine accuracy, something below a certain threshold is the same as zero for a computer, and a number greater than another threshold is treated as infinity (or causes overflow). Moreover, if an equation is not invariant under unit changes, different numerical results will be found for different sets of units.

The usefulness of many estimates can be seriously questioned if this estimate is not dimensionally consistent. For example, the sharpness of the classical Young's inequality

$$\{eq:young1\} \quad ab \leq \frac{a^p}{p} + \frac{b^q}{q}, \quad a, b > 0, \quad \frac{1}{p} + \frac{1}{q} = 1, \quad 1 \leq p, q \leq \infty \text{ (} p \text{ and } q \text{ conjugate)} \quad (1.4)$$

depends on the units chosen if a^p and b^q do not have the same units. Consider for example $p = q = 2$, a velocity and b time. If $a = 1$ km/hour and $b = 1$ hour, $2ab = 2$ km ≤ 1 km²/hour² + 1 hour² = 2 mixed units, but the inequality is very sharp (it is an equality). However, if we decide to change the time units to seconds, we would have $a = 1/3600$ km/s, $b = 3600$ s, $2ab = 2$ km $\leq 3600^{-2}$ km²/s² + 3600² s² = 1.296×10^7 mixed units. Obviously, numerically the inequality $2ab \leq a^2 + b^2$ always holds, but it is useless to get a sharp bound for ab . In this case, the remedy is simple: instead of Eq. (1.4) we may use

$$\{eq:young2\} \quad ab \leq \frac{\alpha^p a^p}{p} + \frac{b^q}{\alpha^q q}, \quad \alpha > 0, \quad (1.5)$$

and choose α to have a dimensionally consistent inequality.

The alternative to consider units is to introduce dimensionless numbers, as many as the number of variables in play minus the number of independent dimensions, according to Buckingham's Theorem. However, this also requires to track the dimensionless numbers in all the analysis and, more importantly, most of the times the way to define them is not unique. Therefore, we will stick to the equations written in dimensional form, although frequent reference will be made to dimensionless numbers.

Since the equations we will deal with arise from basic physics (and in particular continuum physics), we will need basic units, namely, mass, length and time, that we shall denote M, L and T, respectively, as well as temperature K. Other units may be introduced when necessary. Likewise, spatial coordinates $\mathbf{x} = (x_1, \dots, x_d)$ will be considered to have units of length L. Thus, spatial derivatives of order n will introduce units L⁻ⁿ.

We will use the symbol $[\cdot]$ to denote a unit. Very often, in Eq. (1.1) one has that $[f] = [u]T^{-1}$, for example when this equation comes from the energy conservation. In this case, $[g] = [u]LT^{-1}$ and $[\sigma] = T^{-1}$, $[\kappa] = L^2T^{-1}$ and $[\beta] = LT^{-1}$.

1.2 Variational form

1.2.1 The formal weak form

The finite element (FE) method has as a point of departure the so called **weak** or **variational form** of the problem, not the **strong** or **differential form** given by Eqs. (1.1)-(1.3). Let us proceed formally to obtain this variational form. Let $v : \Omega \rightarrow \mathbb{R}$ be any function, with regularity to be determined, with the sole condition that $v = 0$ on Γ_u (for reasons to be clear later on). Multiplying both sides of Eq. (1.1) by v , integrating over Ω and using Gauss' divergence theorem, we obtain:

$$\begin{aligned} \int_{\Omega} v f &= -\kappa \int_{\Omega} v \nabla \cdot \nabla u + \sigma \int_{\Omega} v u \\ &= -\kappa \int_{\Omega} \nabla \cdot (v \nabla u) + \kappa \int_{\Omega} \nabla v \cdot \nabla u + \sigma \int_{\Omega} v u \\ &= -\kappa \int_{\partial\Omega} v \mathbf{n} \cdot \nabla u + \kappa \int_{\Omega} \nabla v \cdot \nabla u + \sigma \int_{\Omega} v u. \end{aligned}$$

Using Eq. (1.3) and recalling that $v = 0$ on Γ_u :

$$\int_{\Omega} v f = \kappa \int_{\Omega} \nabla v \cdot \nabla u - \int_{\Gamma_g} v g + \beta \int_{\Gamma_g} v u + \sigma \int_{\Omega} v u.$$

Therefore, if u solves Problem (1.1)-(1.3), there holds

$$\kappa \int_{\Omega} \nabla v \cdot \nabla u + \sigma \int_{\Omega} v u + \beta \int_{\Gamma_g} v u = \int_{\Omega} v f + \int_{\Gamma_g} v g \quad \forall v \text{ such that } v = 0 \text{ on } \Gamma_u. \quad (1.6) \quad \{\text{eq:c1v1}\}$$

In fact, to obtain this equation we have used Eqs. (1.1) and (1.3), but not Eq. (1.2). If we wish to recover Problem (1.1)-(1.3) from Eq. (1.6), this boundary condition needs to be explicitly required. Eq. (1.6) is the weak or variational statement corresponding to Eqs. (1.1) and (1.3). The objective now is to formalise this variational form and to establish its equivalence with the differential form.

1.2.2 Some functional spaces

Let us recall some definitions to determine the functional environment to pose the problem:

- A metric vector space (not necessarily finite dimensional) is said to be **complete** if it contains the limits of all Cauchy sequences (recall that a sequence $\{x_n\}$ is a Cauchy sequence if for all $\epsilon > 0$ there exists $N \in \mathbb{N}$ such that $\text{dist}(x_n, x_m) < \epsilon$ for $n, m > N$, dist being the metric).
- A complete normed vector space is called a **Banach space**. The norm of a Banach space X will be represented as $\|\cdot\|_X$.
- A complete Euclidean vector space is called a **Hilbert space**. The inner product of a Hilbert space X will be represented as $(\cdot, \cdot)_X$.
- A Hilbert space H is **separable** if there exists a countable set that spans a space dense in H .

Given a region ω of $\mathbb{R}^d$, which can be a surface or a curve, we will make extensive use of the **Lebesgue spaces**:

$$L^p(\omega) := \left\{ f : \omega \rightarrow \mathbb{R} \mid \|f\|_{L^p(\omega)} := \left(\int_{\omega} |f|^p \right)^{\frac{1}{p}} < \infty \right\} \quad \text{for } 1 \leq p < \infty,$$

$$L^\infty(\omega) := \left\{ f : \omega \longrightarrow \mathbb{R} \mid \|f\|_{L^\infty(\omega)} := \operatorname{ess\,sup}_{x \in \omega} |f(x)| < \infty \right\}.$$

It can be shown that $\|\cdot\|_{L^p(\omega)}$ is a norm that endows $L^p(\omega)$ of a Banach space structure, $1 \leq p \leq \infty$. For the particular case $p = 2$, $L^2(\omega)$ can be endowed a Hilbert space structure with the inner product:

$$(f, g)_{L^2(\omega)} := \int_{\omega} f g, \quad f, g \in L^2(\omega).$$

When $\omega = \Omega$, we will simply write $\|\cdot\|_{L^2(\Omega)} \equiv \|\cdot\|$ and $(\cdot, \cdot)_{L^2(\Omega)} \equiv (\cdot, \cdot)$.

In general, given two functions f, g such that their product is integrable, we shall write

$$\langle f, g \rangle_{\omega} := \int_{\omega} f g,$$

which obviously will coincide with the $L^2(\omega)$ -inner product when $f, g \in L^2(\omega)$. Again, when $\omega = \Omega$ we shall write $\langle \cdot, \cdot \rangle_{\Omega} \equiv \langle \cdot, \cdot \rangle$.

Given a Banach space X of functions defined in a region ω , we will define duality with respect to the integral. Thus, the topological dual X' of X will be the set of functions g such that $\langle f, g \rangle_{\omega} < \infty$ for all $f \in X$, with the corresponding dual topology. To emphasise that in this case $\langle f, g \rangle_{\omega}$ is a duality, we may write it as $\langle f, g \rangle_{X \times X'}$. Of course other dualities could be considered, but the one based on the integral is what we will use in these notes. The norm in X' is defined as

$$\|g\|_{X'} = \sup_{f \in X, f \neq 0} \frac{\langle f, g \rangle_{X \times X'}}{\|f\|_X},$$

and thus we have that

$$\langle f, g \rangle_{\omega} \leq \|f\|_X \|g\|_{X'}, \quad f \in X, g \in X'.$$

If $X = L^p(\omega)$, $1 < p < \infty$, it turns out that $X' = L^q(\omega)$, with p and q conjugate, and Hölder's inequality holds:

$$\|fg\|_{L^1(\omega)} \leq \|f\|_{L^p(\omega)} \|g\|_{L^q(\omega)}, \quad f \in L^p(\omega), g \in L^q(\omega).$$

When $p = q = 2$, this reduces to Schwarz's inequality:

$$(f, g)_{L^2(\omega)} \leq \|f\|_{L^2(\omega)} \|g\|_{L^2(\omega)}, \quad f, g \in L^2(\omega).$$

We will also need the **Sobolev spaces**

$$W^{m,p}(\omega) := \{f : \omega \longrightarrow \mathbb{R} \mid D^\alpha f \in L^p(\omega) \forall \alpha \mid |\alpha| \leq m\}, \quad m = 0, 1, 2, \dots, 1 \leq p \leq \infty.$$

Here, $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}^d$ is a multi-index, with $|\alpha| = \alpha_1 + \dots + \alpha_d$ and

$$D^\alpha f := \frac{\partial^{|\alpha|} f}{\partial x_1^{\alpha_1} \dots \partial x_d^{\alpha_d}}.$$

These spaces have a Banach structure with the norm

$$\|f\|_{W^{m,p}(\omega)} := \sum_{|\alpha| \leq m} \ell_\omega^{|\alpha|} \|D^\alpha f\|_{L^p(\omega)}.$$

Here, ℓ_ω represents a characteristic length scale of the domain ω . It is not common to introduce it (although not completely unusual, either), but we have chosen to include it to have a dimensionally consistent sum. We will also need the seminorm

$$|f|_{W^{m,p}(\omega)} := \sum_{|\alpha|=m} \|D^\alpha f\|_{L^p(\omega)}.$$

When $p = 2$, we shall write

$$H^m(\omega) \equiv W^{m,2}(\omega).$$

These are Hilbert spaces with the inner product

$$(f, g)_{H^m(\omega)} := \sum_{|\alpha| \leq m} \ell_\omega^{2|\alpha|} (D^\alpha f, D^\alpha g)_{L^2(\omega)}.$$

Functions in $H^1(\omega)$ turn out to have a well defined trace on $\partial\omega$. If $\gamma \subset \partial\omega$, we define:

$$H^{1/2}(\gamma) := \left\{ f : \gamma \longrightarrow \mathbb{R} \mid \exists \tilde{f} \in H^1(\omega) \text{ with } \tilde{f}|_\gamma = f \right\},$$

$\tilde{f}|_\gamma$ standing for the trace of $\tilde{f}$ on γ . Function $\tilde{f}$ is the extension of f to ω . $H^{1/2}(\gamma)$ is a Banach space with norm

$$\|f\|_{H^{1/2}(\gamma)} = \inf \{ \|\tilde{f}\|_{H^1(\omega)} \mid \tilde{f}|_\gamma = f \}.$$

It is not difficult to see that $H^{1/2}(\gamma) \subset L^2(\gamma)$ and $\ell_\gamma^{1/2} \|f\|_{L^2(\gamma)} \leq \|f\|_{H^{1/2}(\gamma)}$.

We will make use of the space:

$$H_0^1(\omega) := \{ f \in H^1(\omega) \mid f = 0 \text{ on } \partial\omega \}.$$

We could define $H^{-m}(\omega)$ as the topological dual of $H^m(\omega)$. However, for $m = 1$ it is convenient to define $H^{-1}(\omega) = (H_0^1(\omega))'$, as in this case for $u \in H_0^1(\omega)$, $-\Delta u$ is a form of $H^{-1}(\omega)$ that satisfies

$$\langle v, -\Delta u \rangle_{H_0^1(\omega) \times H^{-1}(\omega)} = \int_\omega v(-\Delta u) = \int_\omega \nabla v \cdot \nabla u = \int_\omega (-\Delta v)u = \langle u, -\Delta v \rangle_{H_0^1(\omega) \times H^{-1}(\omega)}.$$

The topological dual of $H^{1/2}(\gamma)$ is denoted as $H^{-1/2}(\gamma)$, although when γ is a curve if $d = 2$ or a surface if $d = 3$ that splits ω with $\partial\gamma \subset \partial\omega$, this symbol is often used for the dual of $H_{00}^{1/2}(\gamma)$, the space of traces on γ of functions in $H_0^1(\omega)$.

The Poincaré-Friedrichs inequality states that for $v \in H_0^1(\omega)$ there holds

$$\|v\|_{L^2(\omega)} \leq C_\omega \|\nabla v\|_{L^2(\omega)}, \tag{1.7} \quad \{\text{eq:c1pf1}\}$$

for a constant C_ω that depends on the domain ω . This has units of length, and in fact it is natural to take $\ell_\omega = C_\omega$. In this case we have that, for $v \in H_0^1(\omega)$,

$$\|v\|_{H^1(\omega)} = \|v\|_{L^2(\omega)} + \ell_\omega \|\nabla v\|_{L^2(\omega)} \leq 2\ell_\omega \|\nabla v\|_{L^2(\omega)} \leq 2\|v\|_{H^1(\omega)}. \tag{1.8} \quad \{\text{eq:c1pf2}\}$$

It is seen that the $L^2(\omega)$ norm of the gradient is equivalent to the $H^1(\omega)$ norm.

Poincaré-Friedrichs' inequality does not only hold for functions that vanish on the whole boundary $\partial\omega$, but also for functions that are zero on part of it. This is enough to control the $L^2(\omega)$ norm of the function if the gradient is already controlled. Its proof in 1D is an interesting exercise, whereas for $d \geq 2$ its proof shows that it also holds if $L^2(\omega)$ is replaced by $L^p(\omega)$, $1 \leq p < \infty$.

1.2.3 The weak form in its functional setting

Let us see which is the functional setting required in Eq. (1.6). We need all terms in this expression to be bounded. Clearly, a sufficient condition for the left-hand-side (LHS) to be bounded is that $u, v \in H^1(\Omega)$, since in this case their traces on Γ_g belong to $H^{1/2}(\Gamma_g) \subset L^2(\Gamma_g)$ and using Schwarz's inequality we have that

$$\int_{\Omega} \nabla v \cdot \nabla u \leq \|\nabla u\| \|\nabla v\| < \infty, \quad \int_{\Omega} vu \leq \|u\| \|v\| < \infty, \quad \int_{\Gamma_g} vu \leq \|u\|_{L^2(\Gamma_g)} \|v\|_{L^2(\Gamma_g)} < \infty.$$

The condition $u, v \in H^1(\Omega)$ is also necessary, since if Eq. (1.6) holds for all v , we could take $v = u - \tilde{u}$, $\tilde{u}$ being an extension of $\bar{u}$ to the whole domain Ω . As $\kappa > 0$, this requires $\|\nabla u\| < \infty$, and likewise for $\|u\| < \infty$, either because $\sigma > 0$ or using Poincaré-Friedrichs's inequality.

Once we have that $u, v \in H^1(\Omega)$, the conditions on the data to have a bounded right-hand-side (RHS) are clear: we need $f \in H^{-1}(\Omega)$ and $g \in H^{-1/2}(\Gamma_g)$.

As it has been said, the Robin boundary condition in Eq. (1.3) has been incorporated in Eq. (1.6), but Eq. (1.2) needs to be explicitly required, and likewise for the condition $v = 0$ on Γ_u . As functions in $H^1(\Omega)$ have traces in $H^{1/2}(\Gamma_u)$, we need $\bar{u} \in H^{1/2}(\Gamma_u)$. The spaces to seek u and to take v are, respectively:

$$V_{\bar{u}} := \{w \in H^1(\Omega) \mid w = \bar{u} \text{ on } \Gamma_u\}, \quad V_0 := \{w \in H^1(\Omega) \mid w = 0 \text{ on } \Gamma_u\}.$$

$V_{\bar{u}}$ is called the **space of trial solutions** and V_0 the **space of test functions**.

We can now introduce the bilinear form:

$$a : V_{\bar{u}} \times V_0 \longrightarrow \mathbb{R} \\ u, v \mapsto a(u, v) := \kappa(\nabla v, \nabla u) + \sigma(v, u) + \beta(v, u)_{L^2(\Gamma_g)}, \quad (1.9)$$

as well as the linear form:

$$l : V_0 \longrightarrow \mathbb{R} \\ v \mapsto l(v) := \langle v, f \rangle + \langle v, g \rangle_{\Gamma_g}. \quad (1.10)$$

The linearity of these forms is obvious. With this, we can state the precise weak or variational form of the problem we are considering:

Given $f \in H^{-1}(\Omega)$, $g \in H^{-1/2}(\Gamma_g)$ and $\bar{u} \in H^{1/2}(\Gamma_u)$, find $u \in V_{\bar{u}}$ such that

$$a(u, v) = l(v) \quad \forall v \in V_0. \quad (1.11)$$

Proposition 1 Problem (1.1)-(1.3) is equivalent to Problem (1.11) if derivatives in the former are understood in the sense of distributions, or if the solution to the latter can be shown to be in $C^2(\Omega) \cap C^1(\bar{\Omega})$.

Proof. Most of the proof is technical, requiring the concept of distribution, and we shall skip it. Let us concentrate on the key point of the equivalence in the case of classical solutions. Clearly, the solution to Problem (1.1)-(1.3) is solution of Problem (1.11), as we have obtained the latter from the former. Let $u \in C^2(\Omega) \cap C^1(\bar{\Omega})$ be a classical solution of Problem (1.11). Since $u \in V_{\bar{u}}$, Eq. (1.2) already holds. From integration by parts it is readily checked that:

$$0 = a(u, v) - l(v)$$

$$= \int_{\Omega} (-\kappa \Delta u + \sigma u - f)v + \int_{\Gamma_g} \left(\kappa \frac{\partial u}{\partial n} + \beta u - g \right) v,$$

for all $v \in \mathcal{C}^2(\Omega) \cap \mathcal{C}^0(\bar{\Omega})$ and vanishing on Γ_u , which is clearly a subset of V_0 . Eqs. (1.1) and (1.3) follow now from the Fundamental Lemma of Calculus of Variations. $\square$

This Proposition serves to understand the different role played by boundary conditions (1.2) and (1.3): whereas the former is explicitly required, the latter is incorporated in the variational statement in Problem (1.11). In the variational formulation presented, the Dirichlet condition (1.2) is called **essential**, whereas the Robin condition (1.3) is called **natural**. It is also said that condition (1.2) is imposed **strongly** and condition (1.3) **weakly**.

The Dirichlet condition (1.2) can be considered homogeneous by performing a simple shift of the unknown. This can be done both in the differential and in the variational formulations of the problem. Let us consider this last one, and let $\tilde{u} \in H^1(\Omega)$ be any extension of $\bar{u} \in H^{1/2}(\Gamma_u)$. Calling $w = u - \tilde{u}$, it turns out that $w = 0$ on Γ_u and

$$\begin{aligned} a(w, v) &= a(u, v) - a(\tilde{u}, v) \\ &= l(v) - \int_{\Omega} v(-\kappa \Delta \tilde{u} + \sigma \tilde{u}) - \int_{\Gamma_g} v \left(\kappa \frac{\partial \tilde{u}}{\partial n} + \beta \tilde{u} \right) \\ &= \langle v, \tilde{f} \rangle + \langle v, \tilde{g} \rangle_{\Gamma_g}, \end{aligned}$$

with $\tilde{f} = f + \kappa \Delta \tilde{u} - \sigma \tilde{u} \in H^{-1}(\Omega)$ and $\tilde{g} = g - \kappa \frac{\partial \tilde{u}}{\partial n} - \beta \tilde{u} \in H^{-1/2}(\Gamma_g)$. Therefore, from now on we will take $u = 0$ on Γ_u , being clear how to treat the non-homogeneous case.

One of the advantages of this approach is that now the two arguments of $a(\cdot, \cdot)$ in Eq. (1.9) belong to the same space $V_{\bar{u}} = V_0 \equiv V$. This in particular allows us to consider interchanging the arguments, and observe that in this case a is **symmetric**, i.e., $a(u, v) = a(v, u)$.

Let us show now that the bilinear form in Eq. (1.9) is **continuous**. In finite dimensional vector spaces, bilinear forms are obviously continuous, but in a general Banach space they are continuous if, and only if, they are **bounded**. In our case we have to show that there exists a constant $N_a > 0$ such that

$$a(u, v) \leq N_a \|u\|_V \|v\|_V. \quad (1.12) \quad \{\text{eq:c1cont}\}$$

The constant N_a is in fact the norm of the bilinear form a , i.e., $N_a = \|a\|_{\mathcal{L}(V \times V; \mathbb{R})}$. In our problem, $\|\cdot\|_V = \|\cdot\|_{H^1(\Omega)}$, and continuity of a is easy to prove:

$$\begin{aligned} a(u, v) &\leq \kappa \|\nabla v\| \|\nabla u\| + \sigma \|v\| \|u\| + \beta \|v\|_{L^2(\Gamma_g)} \|u\|_{L^2(\Gamma_g)} \\ &\leq \left(\frac{\kappa}{\ell_{\Omega}^2} + \sigma + \frac{\beta}{\ell_{\Gamma_g}} \right) \|v\|_{H^1(\Omega)} \|u\|_{H^1(\Omega)}, \end{aligned}$$

that is to say,

$$N_a = \frac{\kappa}{\ell_{\Omega}^2} + \sigma + \frac{\beta}{\ell_{\Gamma_g}}.$$

Note that all terms in N_a have units T^{-1} in the case $[f] = [u]\text{T}^{-1}$ and $[g] = [u]\text{LT}^{-1}$.

Likewise, the linear form l in Eq. (1.10) is continuous: there exists a constant $N_l > 0$ such that

$$l(v) \leq N_l \|v\|_V. \quad (1.13) \quad \{\text{eq:c1cont}\}$$

This is immediate to check:

$$l(v) \leq \|v\|_{H^1(\Omega)} \|f\|_{H^{-1}(\Omega)} + \|v\|_{H^{1/2}(\Gamma_g)} \|g\|_{H^{-1/2}(\Gamma_g)}$$

$$\leq \left(\|f\|_{H^{-1}(\Omega)} + \|g\|_{H^{-1/2}(\Gamma_g)} \right) \|v\|_{H^1(\Omega)},$$

that is to say,

$$N_l = \|f\|_{H^{-1}(\Omega)} + \|g\|_{H^{-1/2}(\Gamma_g)}.$$

The constant N_l is the norm of l , i.e., $N_l = \|l\|_{\mathcal{L}(V;\mathbb{R})} \equiv \|l\|_{V'}$. In our problem, if $[f] = [u]\mathbf{T}^{-1}$ and $[g] = [u]\mathbf{L}\mathbf{T}^{-1}$, we have that $[N_l] = [u]\mathbf{L}^{d/2}\mathbf{T}^{-1}$.

The last property we wish to check of the bilinear form a we are considering is that it is **coercive**, i.e., there exists a constant $K_a > 0$ such that

$$a(v, v) \geq K_a \|v\|_V^2, \quad \forall v \in V. \quad (1.14)$$

This is trivially checked for a given in Eq. (1.9):

$$a(v, v) \geq \kappa \|\nabla v\|^2 + \sigma \|v\|^2 + \beta \|v\|_{L^2(\Gamma_g)}^2 \geq \frac{\kappa}{4\ell_\Omega^2} \|u\|_{H^1(\Omega)}^2,$$

where we have used Eq. (1.8). We see thus that

$$K_a = \frac{\kappa}{4\ell_\Omega^2}.$$

We have not considered the contribution of σ and β to K_a because these parameters can be zero.

The following abstract results for coercive problems obviously apply to the Poisson-type problem we are considering.

1.2.4 Basic theory of coercive problems

Let V a Hilbert space, $a : V \times V \longrightarrow \mathbb{R}$ a bilinear form and $l : V \longrightarrow \mathbb{R}$ a linear form, and consider the problem: find $u \in V$ such that

$$a(u, v) = l(v) \quad \forall v \in V. \quad (1.15)$$

Theorem 1 (Lax-Milgram's lemma) *Suppose that V is separable, the bilinear form a in Eq. (1.15) is continuous and coercive, and that the linear form l is continuous. Then, this equation has a unique solution $u \in V$ that is bounded as*

$$\|u\|_V \leq \frac{N_l}{K_a}. \quad (1.16)$$

Proof. Let us start showing that if the solution exists, it is unique. Suppose that there are two solutions, $u_1, u_2 \in V$. Then:

$$a(u_1, v) = a(u_2, v) = l(v) \quad \forall v \in V \Rightarrow a(u_1 - u_2, v) = 0 \quad \forall v \in V.$$

Taking $v = u_1 - u_2$ and using the coercivity of a :

$$0 = a(u_1 - u_2, u_1 - u_2) \geq K_a \|u_1 - u_2\|_V^2 \Rightarrow u_1 = u_2.$$

There are several ways to prove existence, the most popular probably being one based on Riesz's Representation Theorem (if a is symmetric, as in our case, the proof is straightforward). However, thinking about the approximation to the problem, let us prove existence using the constructive Galerkin method. It requires V to be separable, which in fact is not needed, but this is not an issue, since in most problems

of interest V is indeed separable. Assume then that $V = \text{span}\{\phi_n\}_{n=1}^\infty$, and let $V_N = \text{span}\{\phi_n\}_{n=1}^N$ be the N -dimensional subspace of V spanned by the first N elements of the basis of V . Consider the problem: find $u_N \in V_N$ such that

$$\{eq:c1gal1\} \quad a(u_N, v_N) = l(v_N) \quad \forall v_N \in V_N. \quad (1.17)$$

Since $\{\phi_n\}_{n=1}^N$ is a basis for V_N , we may write $u_N = \sum_{n=1}^N \alpha_n \phi_n$, and the unknown coefficients $\{\alpha_n\}_{n=1}^N$ will be the solution of the linear system:

$$\sum_{n=1}^N a(\phi_n, \phi_m) \alpha_n = l(\phi_m) \quad m = 1, 2, \dots, N.$$

Since a is coercive, the $N \times N$ matrix of this system is invertible, as

$$\sum_{n,m=1}^N a(\phi_n, \phi_m) \alpha_n \alpha_m = a(u_N, u_N) = 0 \Rightarrow u_N = 0 \Rightarrow \alpha_n = 0, \quad n = 1, 2, \dots, N.$$

Thus, there exists a unique solution to Problem (1.17), which satisfies:

$$K_a \|u_N\|_V^2 \leq a(u_N, u_N) = l(u_N) \leq N_l \|u_N\|_V \Rightarrow \|u_N\|_V \leq \frac{N_l}{K_a}.$$

This way we have constructed a bounded sequence $\{u_N\}_{N=1}^\infty$ in V , which must converge weakly to an element $u^* \in V$, i.e., $u_N \rightharpoonup u^*$ as $N \rightarrow \infty$. For M, N fixed, with $1 \leq M \leq N$, there holds $a(u_N, v_M) = l(v_M)$ for all $v_M \in V_M$, and the continuity of a allows one to take the limit in the first argument and arrive to $a(u^*, v_M) = l(v_M)$ for all $v_M \in V_M$ and for all $M \geq 1$. The density of $\{\phi_M\}_{M=1}^\infty$ in V implies that u^* is the solution of Problem (1.15) and satisfies the bound (1.16). $\square$

The previous proof poses the question of how close is u_N to u for a fixed N , i.e., which is the *error* made by *approximating* u by u_N . The answer is given by the next result, which relies on a trivial observation: if Eq. (1.15) holds for all v in V , it also holds for all $v_N \in V_N \subset V$. Subtracting Eq. (1.15) and Eq. (1.17) we get that

$$a(u - u_N, v_N) = 0 \quad \forall v_N \in V_N. \quad (1.18) \quad \{eq:c1-gal1\}$$

This property is often called **Galerkin orthogonality** or, when considering a numerical method, **consistency**. We shall come back to it when considering the FE approximation to the problem.

Theorem 2 (Céa's Lemma) *Under the assumptions of Theorem 1, let $V_N \subset V$. Then, the solution u_N of Problem (1.17) satisfies:*

$$\|u - u_N\|_V \leq \frac{N_a}{K_a} \inf_{v_N \in V_N} \|u - v_N\|_V. \quad (1.19) \quad \{eq:c1-bes\}$$

Proof. Since a is coercive and continuous, we have, for any $v_N \in V_N$:

$$\begin{aligned} K_a \|u - u_N\|_V^2 &\leq a(u - u_N, u - u_N) + a(u - u_N, u_N - v_N) \\ &= a(u - u_N, u - v_N) \\ &\leq N_a \|u - u_N\|_V \|u - v_N\|_V. \end{aligned}$$

In the first step we have used that $a(u - u_N, u_N - v_N) = 0$ because of Eq. (1.18). The result follows from the fact that $v_N \in V_N$ is arbitrary. $\square$

Estimate (1.19) is a *best approximation property*. The distance from u to V_N is $\inf_{v_N \in V_N} \|u - v_N\|_V$ and, according to this estimate, the error $\|u - u_N\|_V$ behaves like this distance, as the constant N_a/K_a is independent of N .

So far we have not used the symmetry of a . If this holds, Problem (1.15) can be shown to correspond to the minimisation of a certain functional, thus linking this variational problem to ‘classical’ variational calculus:

Proposition 2 *Under the assumptions of Theorem 1, suppose also that the bilinear form a is symmetric. Then, the solution $u \in V$ of Problem (1.15) minimises the functional:*

$$J(v) := \frac{1}{2}a(v, v) - l(v)$$

over space V .

Proof. Let $u \in V$ be the solution of Problem (1.15) and fix $v \in V$ arbitrary. Consider the function $f(\epsilon) = J(u + \epsilon v)$. We need to check that f has a minimum at $\epsilon = 0$, and this is trivially verified using the symmetry and the coercivity of a :

$$\begin{aligned} f'(\epsilon)|_{\epsilon=0} &= \frac{1}{2}a(u, v) + \frac{1}{2}a(v, u) - l(v) = 0 \iff a(u, v) = l(v), \\ f''(\epsilon) &= a(u, u) \geq K_a \|u\|_V^2 > 0 \text{ for } u \neq 0. \end{aligned}$$

The theorem follows considering that v is arbitrary. □

Even though considering Problem (1.15) suffices, sometimes it is convenient to write it in operator form. Note that, for $u \in V$ given, $a(u, \cdot)$ is a form of V' defined by $v \mapsto a(u, v)$. Thus, we can define an operator $A : V \longrightarrow V'$ by $u \mapsto Au := a(u, \cdot) \in V'$, with $\langle Au, v \rangle_{V' \times V} = a(u, v)$. Likewise, $\langle f, \cdot \rangle$ is a form of V' , still denoted by f , and we may write Eq. (1.15) as

$$Au = f \quad \text{in } V'.$$

In our case, A can be understood as the operator $-\kappa\Delta + \sigma I$ incorporating the boundary conditions.

1.3 Conforming finite element approximation

The FE method can be understood as a particular way to construct a subspace $V_N \subset V$, which now we will represent as V_h , with $\dim V_h < \infty$. The solution will be sought in this space and the test functions will also be taken in it. Thus, C  a’s Lemma can be applied and we will have a best approximation property guaranteed.

It is important to stress that in what follows we consider $V_h \subset V$. When the approximating space is a subspace of the space of the continuous problem, we call the approximation **conforming**. This will *not* hold for the discontinuous Galerkin method explained later.

The essence of the FE theory is to construct the space V_h and to give an estimate for the **interpolation error**:

$$E_{\text{int}}(h, u) := \inf_{v_h \in V_h} \|u - v_h\|_V, \tag{1.20}$$

for any function $u \in V$. In the problem we consider now, and for u the solution to this problem, when used in Eq. (1.19) this will yield an **a priori estimate** for the error of the FE solution, which will be denoted $u_h \in V_h$.

It is observed that $E_{\text{int}}(h, u)$ is nothing but the distance from u to V_h measured with the metric induced by the norm $\|\cdot\|_V$.

We will not develop the basic FE theory. We will just describe some results that hopefully allow one to understand the general results that we will state. However, it would be highly desirable for the reader to consult general books about the mathematical theory of the FE method.

1.3.1 Construction of finite element spaces

Finite element partition

The FE method starts with the discretisation of the computational domain Ω . For reasons to be explained in the following, when $d = 1$, Ω is split into segments, when $d = 2$ into triangles or quadrilaterals and when $d = 3$ into tetrahedra, hexahedra or prisms with faces that are either triangles or quadrilaterals. In all cases, we shall assume that the vertices of one subdomain either belong to $\partial\Omega$ or are shared by another subdomains. Vertices of one subdomain on the boundary of the other are not allowed, although we will comment later on this possibility. Other generalisations are possible, but we will not consider them here.

The discretisation of the domain is supposed to be a **partition**, i.e., the union of all subdomains is Ω and the intersection of their interiors is empty. This poses the question about the shape of the subdomains and the matching with Ω . We will consider that the boundaries of the subdomains can be represented by polynomials to be specified below and that the union of all of them exactly matches Ω . Of course, in the applications this does not necessarily happen, and therefore there is an inherent approximation to Ω ; neglecting this approximations is one of **Strang's variational crimes**.

Let $\mathcal{T}_h = \{K\}$ be the partition of Ω , assumed to be made of n_{ele} subdomains. Here and below it is understood that K is a generic subdomain of the partition and that $\{K\}$ is the collection of all such subdomains. It is assumed that operations indexed by K apply to all subdomains. There will hold that $\bar{\Omega} = \cup_K \bar{K}$, and that $\kappa_1 \cap \kappa_2 = \emptyset$ for $K_1, K_2 \in \mathcal{T}_h$, $K_1 \neq K_2$. We define:

$$h_K := \text{diam}(K), K \in \mathcal{T}_h, \quad h = \max_{K \in \mathcal{T}_h} h_K.$$

h_K is the diameter of subdomain K (minimum diameter of the balls that contain K , i.e., circles if $d = 2$ or spheres if $d = 3$) and h is referred to as the **diameter of the FE partition** $\mathcal{T}_h$. The dimension of the FE spaces to be constructed depends on h , growing as h decreases, and therefore the FE approximation has to be analysed when $h \rightarrow 0$. Thus, we have to think of a family of FE partitions $\{\mathcal{T}_h\}_{h>0}$. Calculations will be done for fixed values of h , but the analysis has to determine the behaviour of the error as $h \rightarrow 0$, in a sense to be explained below.

Increasing the dimension of V_h by reducing h , or at least some of the h_K 's, is called **h -refinement**. However, there are different ways to perform this refinement. The simplest is the **uniform** refinement, in which h_K is decreased by the same factor for all $K \in \mathcal{T}_h$. This is the case for example when all subdomains are equally subdivided into smaller subdomains.

Let us define ρ_K as the maximum diameter of the balls contained in $K \in \mathcal{T}_h$. If there exists $\gamma_{\max} < \infty$ such that

$$\gamma_K = \frac{h_K}{\rho_K} < \gamma_{\max}, \quad \forall K \in \mathcal{T}_h, \forall h > 0, \quad (1.21) \quad \{\text{eq:c1-iso}\}$$

the family $\{\mathcal{T}_h\}_{h>0}$ is called **shape regular**. The parameter γ_K is called **aspect ratio** of element K . Let $h_{\min} = \min_{K \in \mathcal{T}_h} (h_K)$. If $\{\mathcal{T}_h\}_{h>0}$ is shape regular and also

$$\frac{h}{h_{\min}} < \infty \quad \forall K \in \mathcal{T}_h, \forall h > 0, \quad (1.22) \quad \{\text{eq:c1-uni}\}$$

the refinement is called **quasi-uniform**.

Most of the results to be presented regarding interpolation estimates and bounds for FE functions will be based on the assumption that the family of FE partitions is quasi-uniform. However, there are important refinements in which either Eq. (1.21) or Eq. (1.22) do not hold.

Condition (1.21) essentially means the the refinement is isotropic, that is to say, all elements decrease their size in all directions at the same rate. However, in many practical situations it is convenient to decrease the size in one direction faster than in the others. In fluid mechanics, this happens for example when trying to capture boundary layers, shear layers or shocks. As a trivial example, consider the unit square $\Omega = (0, 1) \times (0, 1)$ divided into quads, with n subdomains in the x -direction and n^2 subdomains in the y -direction. Then, $h_K = \frac{1}{n}$ and $\rho_K = \frac{1}{n^2}$ for all K , and $\frac{h_K}{\rho_K} = n \rightarrow \infty$ as $n \rightarrow \infty$, i.e., as $h \rightarrow 0$. Refinements that do not fulfil condition (1.21) are called **anisotropic**. Even though they are very important in the applications, the theory about them is not fully complete. We will not consider them in these notes.

The meaning of condition (1.22) is that all elements decrease their diameter at the same rate as $h \rightarrow 0$. However, in the applications it is very common that the refinement is only needed in certain regions of the computational domain, not everywhere. As we shall see, the interpolation estimates depend on derivatives of the function to be interpolated, and if these derivatives are different at different points, it makes sense to use different refinements in different regions. This is the essence of the so called **local mesh refinement**, tightly linked to the concept of **adaptivity**, i.e., the selection of different mesh sizes for different regions of the computational domain. The way to do this depends on the availability of **a posteriori error estimates**, that is to say, estimates of the error of the FE solution once it has been computed. This is also an important subject that will not be treated in these notes.

Associated to the FE partition $\mathcal{T}_h$ we will also need the collection of edges $\mathcal{E}_h = \{E\}$. If $d = 2$ these will be segments, whereas if $d = 3$ they will be faces. Nevertheless, we shall always employ the term **edge**. A boundary edge will be of the form $E = \partial K \cap \partial\Omega$, whereas an interior edge will be

$$E = \partial K_1 \cap \partial K_2, \quad K_1, K_2 \in \mathcal{T}_h.$$

Recall that we assume that the vertices of K_1 on ∂K_2 are also vertices of K_2 .

Polynomial interpolation

Perhaps the most distinctive feature of the FE method is that it is based on piecewise polynomial approximations. The key idea is to consider that V_h is made of polynomials in each subdomain K of the FE partition. This idea is elaborated in many textbooks; here we will directly describe the version that is usually implemented in FE codes, based on defining the polynomial interpolation in a reference subdomain, which we will call **reference element**.

Suppose that K is a triangle or a quad if $d = 2$ and a tetrahedron or a hexahedron if $d = 3$; the treatment of other prisms is straightforward from what follows. This subdomain is defined in the Cartesian space parametrised with coordinates $\mathbf{x} = (x_1, \dots, x_d)$. We define in each case a reference element $\hat{K}$, parametrised by the Cartesian coordinates $\boldsymbol{\xi} = (\xi_1, \dots, \xi_d)$, and defined by

$$\hat{K} = \begin{cases} \{\boldsymbol{\xi} \in \mathbb{R}^d \mid 0 \leq \xi_i \leq 1, i = 1, \dots, d, \quad \xi_1 + \dots + \xi_d \leq 1\} & \text{if } K \text{ is a simplex,} \\ \{\boldsymbol{\xi} \in \mathbb{R}^d \mid -1 \leq \xi_i \leq 1, i = 1, \dots, d\} & \text{if } K \text{ is a quad or an hexahedron.} \end{cases}$$

On $\hat{K}$ we define the set of polynomials

$$R_k(\hat{K}) = \begin{cases} P_k(\hat{K}) & \text{if } K \text{ is a simplex,} \\ Q_k(\hat{K}) & \text{if } K \text{ is a quad or an hexahedron,} \end{cases}$$

where

$$\begin{aligned} P_k(\hat{K}) &= \{p : \hat{K} \rightarrow \mathbb{R} \mid p(\boldsymbol{\xi}) \text{ is a polynomial of degree } k\}, \\ Q_k(\hat{K}) &= \{p : \hat{K} \rightarrow \mathbb{R} \mid p(\boldsymbol{\xi}) \text{ is a polynomial of degree } k \text{ in each coordinate}\}. \end{aligned}$$

Let us consider now a diffeomorphism

$$\begin{aligned} \varphi_K : \hat{K} &\longrightarrow K \\ \boldsymbol{\xi} &\mapsto \mathbf{x} = \varphi_K(\boldsymbol{\xi}). \end{aligned} \tag{1.23} \quad \{\text{eq:c1-dif}\}$$

We can finally construct the approximation space:

$$R_{h,k} = \left\{ v_h : \Omega \rightarrow \mathbb{R} \mid v_h|_K = \hat{v} \circ \varphi_K^{-1}, \hat{v} \in R_k(\hat{K}), K \in \mathcal{T}_h \right\}. \tag{1.24} \quad \{\text{eq:c1-aps}\}$$

This will be the basic space to construct FE spaces. Sometimes it is called **broken FE space**. Functions in $R_{h,k}$ are defined on each subdomain $K \in \mathcal{T}_h$, and therefore not necessarily continuous. When transformed (pulled back) to the reference domain $\hat{K}$ they are complete polynomials of degree k , and therefore when considered in the subdomain K they are not necessarily polynomials. However, we still need to define φ_K . This diffeomorphism is also taken to be a polynomial of degree k' , i.e., $\varphi_K \in R_{k'}(\hat{K})^d$. If $k' = k$, the representation of K from $\hat{K}$ is said to be **isoparametric**. If $k' < k$ (with the obvious restriction $k' \geq 1$) the representation is called **subparametric**, and **superparametric** if $k' > k$. In practice, either $k' = 1$ or isoparametric representations are used. If $k' = 1$, the mapping φ_K will be affine, and therefore $v_h|_K = \hat{v} \circ \varphi_K^{-1}$ will be a polynomial of degree k in K , i.e., we may write $v_h|_K \in R_k(K)$.

For $k' > 1$, φ_K allows one to represent elements with curved boundaries. For $E \subset \partial\Omega$, $E \in \mathcal{E}_h$, this may serve to approximate better the shape of Ω (recall that we will not analyse this approximation). However, there is no need (in general) to use curved interior edges, and moreover, this may induce conformity issues. Indeed, for $k' > 1$ and E curved, $\hat{v} \circ \varphi_K^{-1}$ is a rational function when restricted to $\varphi_K^{-1}(E)$ ($E \subset \partial K$), and this may complicate for example ensuring continuity between adjacent elements; we shall come back later to this point. As a conclusion, interior edges will be considered affine with the edges of the reference element.

The set $R_{h,k}$ in Eq. (1.24) will be the basis to construct the FE spaces for scalar functions. In the case of vector fields, different component can be approximated in different ways. Moreover, so far we have posed no continuity requirement between subdomains of the FE partition. Let us finally note that we have assumed the same polynomial degree k for all $K \in \mathcal{T}_h$, but it is trivial to extend the definition of $R_{h,k}$ to consider $k = k(K)$, i.e., a different polynomial order for each subdomain K .

Degrees of freedom

Once $R_{h,k}$ is defined, the next question is how to represent polynomial functions $\hat{v}(\boldsymbol{\xi})$ in the reference domain $\hat{K}$ or, equivalently, which is the basis chosen for $R_k(\hat{K})$. Observe that

$$\dim P_k(\hat{K}) = \binom{d+k}{d} = \frac{(d+k)!}{d!k!}, \quad \dim Q_k(\hat{K}) = (k+1)^d.$$

Let $n_{\text{ino}} = \dim R_k(\hat{K})$ be either of these two dimensions. If we choose a basis for this space, so that

$$R_k(\hat{K}) = \text{span}\{\hat{N}^1(\boldsymbol{\xi}), \dots, \hat{N}^{n_{\text{ino}}}(\boldsymbol{\xi})\}, \tag{1.25} \quad \{\text{eq:c1-sha}\}$$

we will be able to write any element $\hat{v}(\boldsymbol{\xi}) \in R_k(\hat{K})$ as

$$\hat{v}(\boldsymbol{\xi}) = \sum_{A=1}^{n_{\text{ino}}} \hat{N}^A(\boldsymbol{\xi}) \alpha^A$$

for a set of coefficients $\{\alpha^A\}_{A=1}^{n_{\text{Ino}}}$. Note that we use a superscript for the basis functions and the coefficients that multiply them, not to be confused with an exponent.

The basis functions in Eq. (1.25) are called **shape functions**. Note that they are defined in $\hat{K}$, their transformation to K being

$$N_K^A(\mathbf{x}) = \hat{N}^A(\varphi_K^{-1}(\mathbf{x})), \quad A = 1, \dots, n_{\text{Ino}}, \quad \mathbf{x} \in K. \quad (1.26)$$

In the way we have constructed it, $N_K^A(\mathbf{x})$ is not necessarily a polynomial, but depends on the expression of φ_K . However, this is the way the shape functions are constructed in practice, as we shall see. In fact, since we have defined them for each $K \in \mathcal{T}_h$, Eq. (1.26) gives the **local** shape functions. Any $v_h \in R_{h,k}$ will be locally given by

$$v_h(\mathbf{x})|_K = \sum_{A=1}^{n_{\text{Ino}}} N_K^A(\mathbf{x}) \alpha_K^A = \sum_{A=1}^{n_{\text{Ino}}} \hat{N}^A(\varphi_K^{-1}(\mathbf{x})) \alpha_K^A.$$

The collection $\{\alpha_K^A\}_{A=1}^{n_{\text{Ino}}}$ is the set of **local degrees of freedom**, which determine any function $v_h \in R_{h,k}$ in subdomain K . They obviously depend on the way the shape functions are chosen. Since given $\hat{v} \in R_k(\hat{K})$ we can construct the mapping $\hat{v} \mapsto \alpha^A$ that for each $\hat{v}$ gives the coefficient of the shape function $\hat{N}^A$, we can think of the degrees of freedom as elements of the dual of $R_k(\hat{K})$.

We are now in a position to define what a **finite element** is: in the reference domain $\hat{K}$ it is defined as the triplet $(\hat{K}, R_k(\hat{K}), \{\alpha^A\}_{A=1}^{n_{\text{Ino}}})$, i.e., a subdomain endowed with a space of polynomials and a set of degrees of freedom (or equivalently, with a basis of the space of polynomials) which has to determine uniquely the elements of $R_k(\hat{K})$ (i.e., the degrees of freedom have to be $R_k(\hat{K})$ -**unisolvant**). This definition can be translated to $K \in \mathcal{T}_h$ by the diffeomorphism φ_K . If any of the ingredient of the triplet changes, the resulting FE will be different. However, abusing of the language one often refers to the subdomains $\{K\}$ as the ‘finite elements’ of the approximation.

All textbooks about the FE method give examples of finite elements at this point. In these notes we will introduce them when needed, but because of its relevance let us define the concept of **Lagrangian** FE. Suppose we are given $\hat{K}$ and $R_k(\hat{K})$. In $\hat{K}$ we pick a set of points $\{\xi^B\}_{B=1}^{n_{\text{Ino}}}$ and choose the basis functions $\{\hat{N}^A(\xi)\}_{A=1}^{n_{\text{Ino}}}$ such that

$$\hat{N}^A(\xi) \in R_k(\hat{K}), \quad \hat{N}^A(\xi^B) = \delta^{AB}, \quad (1.27)$$

where δ^{AB} is the Krönecher delta. Thus, $\hat{N}^A(\xi)$ is a polynomial of degree k that takes the value 1 at ξ^A and 0 at ξ^B , $B \neq A$. Of course, the points $\{\xi^B\}_{B=1}^{n_{\text{Ino}}}$ have to be chosen so that the shape functions can be uniquely determined. These points are called **nodes** of the element. The selection of the shape functions satisfying Eq. (1.27) has as a consequence that *the degrees of freedom of the element will be the nodal values of the function to be represented*. Indeed, if we call $\hat{v}^A = \hat{v}(\xi^A)$ we will have that

$$\hat{v}(\xi) = \sum_{A=1}^{n_{\text{Ino}}} \hat{N}^A(\xi) \hat{v}^A. \quad (1.28)$$

This is trivially checked by noting that both sides of this expression are polynomials of $R_k(\hat{K})$ that coincide at the nodes $\{\xi^B\}_{B=1}^{n_{\text{Ino}}}$, and these have been chosen to guarantee that the bases are uniquely determined.

The basis in subdomain $K \in \mathcal{T}_h$ will be simply determined by φ_K as $N_K^A(\mathbf{x}) = \hat{N}^A(\varphi_K^{-1}(\mathbf{x}))$. However, in this case there is a natural choice for the mapping φ_K . If we consider the isoparametric representation and express the components of φ_K in terms of the Lagrangian basis, we can construct:

$$\mathbf{x}|_K = \varphi_K(\xi) = \sum_{A=1}^{n_{\text{Ino}}} \hat{N}^A(\xi) \mathbf{x}_K^A,$$

where $\mathbf{x}_K^A$ is the transformed of node A in the parent domain $\hat{K}$ to K . Indeed, by virtue of Eq. (1.27) we will have that

$$\mathbf{x}^B|_K = \varphi_K(\boldsymbol{\xi}^B) = \sum_{A=1}^{n_{\text{ino}}} \hat{N}^A(\boldsymbol{\xi}^B) \mathbf{x}_K^A = \mathbf{x}_K^B, \quad B = 1, \dots, n_{\text{ino}}.$$

Expression (1.28) has its counterpart in K :

$$v_h(\mathbf{x})|_K = \sum_{A=1}^{n_{\text{ino}}} N_K^A(\mathbf{x}) v_K^A, \quad (1.29) \quad \{\text{eq:c1-exn}\}$$

where $v_K^A = v_h(\mathbf{x}_K^A)$. This results from the fact that

$$N_K^A(\mathbf{x}_K^B) = \hat{N}^A(\varphi_K^{-1}(\mathbf{x}_K^B)) = \hat{N}^A(\boldsymbol{\xi}^B) = \delta^{AB}.$$

The expression of the shape functions $\{\hat{N}^A(\boldsymbol{\xi})\}_{A=1}^{n_{\text{ino}}}$ for different Lagrangian elements can be found in any FE textbook. In general, Lagrangian elements constructed on triangles or tetrahedra with $P_k(\hat{K})$ polynomials will be referred to as **P_k -elements**, whereas Lagrangian elements constructed on quads or hexahedra with $Q_k(\hat{K})$ polynomials will be called **Q_k -elements**.

The finite element space for Poisson's problem

The previous general construction of the approximation space in Eq. (1.24) allows us to construct particular FE spaces for the problem at hand. Let us discuss now which should be the correct space to seek the FE solution in the case of Poisson's problem. The key in this case is that if we wish to consider conforming approximations, with $V_h \subset V \subset H^1(\Omega)$, V_h needs to be made of continuous functions. This can be justified in abstract by using a Sobolev's embedding theorem, but in this case it is trivial to see that continuity is a necessary and sufficient condition for functions in $V_h \subset R_{h,k}$ to be in $H^1(\Omega)$. First, if $v_h \in \mathcal{C}^0(\bar{\Omega}) \cap R_{h,k}$ it is clear that its derivatives will be (in general) discontinuous, but square integrable, and thus $v_h \in H^1(\Omega)$. Secondly, if $v_h \notin \mathcal{C}^0(\bar{\Omega})$ its derivatives will behave as Dirac δ -distributions, which are not square integrable; thus, $v_h \notin H^1(\Omega)$. Therefore:

$$V_h = \{v_h \in \mathcal{C}^0(\bar{\Omega}) \cap R_{h,k} \mid v_h = 0 \text{ on } \Gamma_u\}. \quad (1.30) \quad \{\text{eq:c1-Poi}\}$$

Any set of degrees of freedom can be used in $R_{h,k}$. To fix ideas, we will consider Lagrangian FEs.

The condition $v_h = 0$ on Γ_u can be imposed only in an approximate way. If we have a boundary edge E that is not fully contained in Γ_u , if all nodal values of v_h of nodes in E are set to zero, v_h will be zero even outside Γ_u , and if one of them is not zero v_h will not be exactly zero on the whole Γ_u . Thus, we will assume that Γ_u is exactly the union of a set of boundary edges and the nodal values at these edges are all fixed to zero.

The key point in the construction of V_h is how to impose continuity. Let $K_1, K_2 \in \mathcal{T}_h$, with $E = \partial K_1 \cap \partial K_2 \neq \emptyset$. Recall that we have assumed E straight (flat, if $d = 3$), affine to an edge of the reference element. Thus, $v_h|_E$ will be a polynomial of degree k , both computed from ∂K_1 and from ∂K_2 . The two expressions will be given in terms of the nodal values, which have to be $R_k(E)$ -unisolvant. Thus, *continuity is ensured if K_1 and K_2 share nodes and nodal values on E* . This reasoning also serves to understand the convenience of considering straight edges, since otherwise coincident nodes and nodal values would not ensure continuity and, therefore, conformity.

Continuous FE are thus constructed placing nodes on the boundaries, guaranteeing $R_k(\partial K)$ -unisolvence, and considering common the nodes and nodal values of adjacent elements. This brings

the need for a global node numbering, since the number of nodes will be smaller than $n_{\text{ino}} \times n_{\text{ele}}$. Thus, let n_{gno} be the total number of nodes of the mesh, and let $\text{ltg}(A, K)$ be the function that yields the global number of node A in element (subdomain) K . For each global node a we can define a **global shape function** $N^a(\mathbf{x})$ by

$$N^a(\mathbf{x}) = \begin{cases} N_K^A(\mathbf{x}) & \text{if } \mathbf{x} \in K \text{ and } a = \text{ltg}(A, K) \\ 0 & \text{otherwise} \end{cases}$$

Note that the support of $N^a(\mathbf{x})$ is the union of all K to which node a belongs.

Because of Eq. (1.29) we now may write the global expression:

$$v_h(\mathbf{x}) = \sum_{a=1}^{n_{\text{gno}}} N^a(\mathbf{x}) v^a, \quad v^a = v_h(\mathbf{x}^a), \quad \mathbf{x} \in \Omega, \quad (1.31)$$

where $\mathbf{x}^a$ are the coordinates of the global node a .

Let us collect the different expressions of FE functions that we have introduced:

$\hat{v}(\boldsymbol{\xi}) = \sum_{A=1}^{n_{\text{ino}}} \hat{N}^A(\boldsymbol{\xi}) \alpha^A$	General expression in $\hat{K}$,
$\hat{v}(\boldsymbol{\xi}) = \sum_{A=1}^{n_{\text{ino}}} \hat{N}^A(\boldsymbol{\xi}) \hat{v}^A$	Lagrangian elements in $\hat{K}$, $\hat{v}^A = \hat{v}(\boldsymbol{\xi}^A)$,
$v_h(\mathbf{x}) _K = \sum_{A=1}^{n_{\text{ino}}} N_K^A(\mathbf{x}) v_K^A$	Lagrangian elements in K , $v_K^A = v_h(\mathbf{x}^A)$,
$v_h(\mathbf{x}) = \sum_{a=1}^{n_{\text{gno}}} N^a(\mathbf{x}) v^a$	Lagrangian elements in Ω , $v^a = v_h(\mathbf{x}^a)$.

The last one is mostly used in the case of continuous FE functions, although it could be applied in general. All functions in V_h defined in Eq. (1.30) can be expressed in this way. In particular, note that if $v_h = 0$ on Γ_u , the nodal values corresponding to nodes in this boundary will be zero.

1.3.2 Discrete variational problem

Now that we have constructed the FE space V_h , the statement of the discrete FE problem is the following: find $u_h \in V_h$ such that

$$a(u_h, v_h) = l(v_h) \quad \forall v_h \in V_h. \quad (1.32)$$

Let us see which is the algebraic system we have to solve. Since u_h and v_h are FE functions, we can express them as

$$u_h(\mathbf{x}) = \sum_{a=1}^{n_{\text{gnf}}} N^a(\mathbf{x}) u^a, \quad v_h(\mathbf{x}) = \sum_{b=1}^{n_{\text{gnf}}} N^b(\mathbf{x}) v^b,$$

where we have considered that the nodes without Dirichlet boundary conditions are the first n_{gnf} , and those prescribed are $n_{\text{gnf}} + 1, \dots, n_{\text{gno}}$. Using these expressions, problem (1.32) yields:

$$\sum_{a,b=1}^{n_{\text{gnf}}} v^b a(N^b, N^a) u^a = \sum_{b=1}^{n_{\text{gnf}}} v^b l(N^b).$$

If we introduce the matrix and arrays:

$$\begin{aligned} \mathbf{u}^T &= (u^1, \dots, u^{n_{\text{gnf}}}), \quad \mathbf{v}^T = (v^1, \dots, v^{n_{\text{gnf}}}), \\ \mathbf{A}^{ab} &= a(N^b, N^a), \quad \mathbf{A} = [\mathbf{A}^{ab}] \in \mathbb{R}^{n_{\text{gnf}} \times n_{\text{gnf}}}, \\ \mathbf{f}^b &= l(N^b), \quad \mathbf{f} = [\mathbf{f}^b] \in \mathbb{R}^{n_{\text{gnf}}}, \end{aligned}$$

we obtain that

$$\mathbf{v}^T \mathbf{A} \mathbf{u} = \mathbf{v}^T \mathbf{f} \quad \forall \mathbf{v} \in \mathbb{R}^{n_{\text{gnf}}} \iff \mathbf{A} \mathbf{u} = \mathbf{f}. \quad (1.33) \quad \{\text{eq:c1-dis}\}$$

Therefore, obtaining the FE solution $u_h \in V_h$ boils down to obtaining its nodal values by solving this system. Traditionally, matrix $\mathbf{A}$ is called the **stiffness matrix** of the problem and array $\mathbf{f}$ the **vector of nodal forces**, even in problems unrelated to mechanics.

1.3.3 Flow of a finite element implementation

Let us explain now which is the basic flow of the computer implementation to arrive to system (1.33), with

$$\mathbf{A}^{ab} = a(N^b, N^a) = \kappa(\nabla N^a, \nabla N^b) + \sigma(N^a, N^b) + \beta(N^a, N^b)_{L^2(\Gamma_g)}, \quad (1.34) \quad \{\text{eq:c1maA}\}$$

$$\mathbf{f}^a = l(N^a) = \langle N^a, f \rangle + \langle N^a, g \rangle_{\Gamma_g}. \quad (1.35) \quad \{\text{eq:c1maf}\}$$

Although there are implementations that do not do this, let us split matrix $\mathbf{A}$ and array $\mathbf{f}$ into their contribution from volume integrals and their contribution from integrals over Γ_g , i.e.,

$$\mathbf{A} = \mathbf{A}_\Omega + \mathbf{A}_{\Gamma_g}, \quad \mathbf{f} = \mathbf{f}_\Omega + \mathbf{f}_{\Gamma_g}.$$

Let us start with the calculation of $\mathbf{A}_\Omega$ and $\mathbf{f}_\Omega$. Because of the additivity of the integral we have that

$$\mathbf{A}_\Omega^{ab} = \sum_K \mathbf{A}_K^{AB}, \quad \mathbf{f}_\Omega^a = \sum_K \mathbf{f}_K^A \quad \text{with } a = \text{ltg}(A, K), \quad b = \text{ltg}(B, K).$$

The calculation of the global arrays from the local (element) ones is called **assembly**. It involves summation over the element domains and injection, as small arrays are injected into larger ones through the mapping $\text{ltg}(\cdot, \cdot)$. Even though the operations are different when applied to matrices or to vectors, we shall use the same notation:

$$\mathbf{A}_\Omega = \mathbb{A}_{K \in \mathcal{T}_h}(\mathbf{A}_K), \quad \mathbf{f}_\Omega = \mathbb{A}_{K \in \mathcal{T}_h}(\mathbf{f}_K).$$

From Eqs. (1.34)-(1.35) we have that:

$$\begin{aligned} \mathbf{A}_K^{AB} &= \kappa \int_K \nabla N_K^A \cdot \nabla N_K^B + \sigma \int_K N_K^A N_K^B \\ &= \kappa \int_{\hat{K}} \frac{\partial \hat{N}^A}{\partial \xi_i} \frac{\partial \xi_i}{\partial x_j} \frac{\partial \hat{N}^B}{\partial \xi_k} \frac{\partial \xi_k}{\partial x_j} |\det(D\varphi_K)| + \sigma \int_{\hat{K}} \hat{N}^A \hat{N}^B |\det(D\varphi_K)| \\ &= \kappa \int_{\hat{K}} [\nabla_\xi \hat{N}^A \cdot D\varphi_K^{-1}] \cdot [\nabla_\xi \hat{N}^B \cdot D\varphi_K^{-1}] |\det(D\varphi_K)| + \sigma \int_{\hat{K}} \hat{N}^A \hat{N}^B |\det(D\varphi_K)| \\ \mathbf{f}_K^A &= \int_K N^A f \\ &= \int_{\hat{K}} \hat{N}^A (f \circ \varphi_K) |\det(D\varphi_K)| \end{aligned}$$

Here we have used basic results of Calculus: the change of variables theorem, the chain rule and the inverse function theorem. Likewise, we have employed Einstein's notation: repeated indexes that run over the number of space dimensions imply summation.

The last ingredient we need for the implementation is that integrals need to be computed numerically. Observe that they are only required in the reference element $\hat{K}$. The numerical integration rule will be determined by a collection of points in $\hat{K}$, $\{\xi^g\}_{g=1}^{n_{\text{int}}}$ and collection of weights $\{w^g\}_{g=1}^{n_{\text{int}}}$ such that for any function F defined on $\hat{K}$:

$$\int_{\hat{K}} F(\xi) \approx \sum_{g=1}^{n_{\text{int}}} F(\xi^g) w^g.$$

We will not discuss possible integration rules in general, but we will refer to them when necessary. Because of the widespread use of Gauss' integration rules, the integration points are often called Gauss points. However, we will avoid this terminology.

We will not take into account the numerical integration and its effect in the analysis of the FE method. In general terms, the numerical integration has to be such that the accuracy of the method is not spoiled by it (in terms of asymptotic order of convergence), although it will have some effect. Neglecting it is another of Strang's variational crimes.

With all this in mind, the implementation of the calculation of the stiffness matrix and vector of nodal forces is as follows:

For each K do:

 Initialise arrays.

 For $g = 1, \dots, n_{\text{int}}$ do:

 Construct the isoparametric mapping and its Jacobian at ξ^g :

$$\begin{aligned} \mathbf{x}^g &= \varphi_K(\xi^g) = \sum_{A=1}^{n_{\text{lno}}} \hat{N}^A(\xi^g) \mathbf{x}^A \\ (D\varphi_K)_{ij} |_{\xi^g} &= \frac{\partial x_i}{\partial \xi_j} \Big|_{\xi^g} = \sum_{A=1}^{n_{\text{lno}}} \frac{\partial \hat{N}^A}{\partial \xi_j} \Big|_{\xi^g} x_i^A, \quad i, j = 1, \dots, d. \end{aligned}$$

 Compute $\det(D\varphi_K) |_{\xi^g}$.

 Invert $D\varphi_K |_{\xi^g}$.

 For $A = 1, \dots, n_{\text{lno}}$ do:

$$\text{Compute } \nabla N_K^A |_{\xi^g} = \nabla_{\xi} \hat{N}^A \cdot D\varphi_K^{-1} |_{\xi^g}.$$

 For $B = 1, \dots, n_{\text{lno}}$ do:

$$\text{Compute } \nabla N_K^B |_{\xi^g} = \nabla_{\xi} \hat{N}^B \cdot D\varphi_K^{-1} |_{\xi^g}.$$

$$\mathbf{A}_K^{AB} \leftarrow \mathbf{A}_K^{AB} + (\kappa \nabla N_K^A |_{\xi^g} \cdot \nabla N_K^B |_{\xi^g} + \sigma N_K^A |_{\xi^g} N_K^B |_{\xi^g}) |\det(D\varphi_K) |_{\xi^g} w^g$$

 End do

$$\mathbf{f}_K^A \leftarrow \mathbf{f}_K^A + N_K^A |_{\xi^g} f(\mathbf{x}^g) |\det(D\varphi_K) |_{\xi^g} w^g$$

 End do

 End do

End do

Let us consider now the contribution to $\mathbf{A}$ and $\mathbf{f}$ from terms integrated over Γ_g . Again, the additivity of the integral yields:

$$\mathbf{A}_{\Gamma_g}^{ab} = \sum_{E \subset \Gamma_g} \mathbf{A}_E^{AB}, \quad \mathbf{f}_{\Gamma_g}^a = \sum_{E \subset \Gamma_g} \mathbf{f}_E^A \quad \text{with } a = \text{ltg}(A, K), \quad b = \text{ltg}(B, K) \quad \text{when } E \subset \partial K.$$

Here the summation extends over all edges that are contained in Γ_g . We can thus introduce the assembly:

$$\mathbf{A}_{\Gamma_g} = \mathbb{A}_{E \in \mathcal{E}_h, E \subset \Gamma_g}(\mathbf{A}_E), \quad \mathbf{f}_{\Gamma_g} = \mathbb{A}_{E \in \mathcal{E}_h, E \subset \Gamma_g}(\mathbf{f}_E).$$

Let $\hat{E} \subset \partial \hat{K}$ be the edge that is transformed to $E \subset K$ through φ_K . We consider now $d = 3$, the case $d = 2$ being simpler. Without loss of generality, assume that ξ_1, ξ_2 are the coordinates in $\hat{E}$ and let φ_E be the mapping φ_K restricted to $\hat{E}$, so that $\mathbf{x} = \varphi_E(\xi_1, \xi_2) \in E$. The vectors tangent to E at each point are $\mathbf{g}_i = \frac{\partial \varphi_E}{\partial \xi_i}$, $i = 1, 2$, and if $d\hat{S}$ is the surface element in $\hat{E}$, the surface element in E is $dS = |\mathbf{g}_1 \times \mathbf{g}_2|d\hat{S}$. Therefore, we have that

$$\begin{aligned} \mathbf{A}_E^{AB} &= \beta \int_E N_K^A N_K^B = \beta \int_{\hat{E}} \hat{N}^A \hat{N}^B |\mathbf{g}_1 \times \mathbf{g}_2| \\ \mathbf{f}_E^A &= \int_E N_K^A g = \int_{\hat{E}} \hat{N}^A (g \circ \varphi_E) |\mathbf{g}_1 \times \mathbf{g}_2| \end{aligned}$$

These integrals are also computed numerically. Let $\{\xi_E^g\}_{g=1}^{n_{\text{int},E}}$ be the position of the integration points and $\{w_E^g\}_{g=1}^{n_{\text{int},E}}$ the weights. The implementation of these calculations is as follows:

For each $E \in \Gamma_g$ do:

 Initialise arrays.

 For $g = 1, \dots, n_{\text{int},E}$ do:

 Construct the isoparametric mapping and the tangent basis at $(\xi_{1,E}^g, \xi_{2,E}^g)$:

$$\mathbf{x}_E^g = \varphi_E(\xi_{1,E}^g, \xi_{2,E}^g) = \sum_{A=1}^{n_{\text{ino}}} \hat{N}^A(\xi_{1,E}^g, \xi_{2,E}^g) \mathbf{x}^A,$$

$$\mathbf{g}_i(\xi_{1,E}^g, \xi_{2,E}^g) = \frac{\partial \varphi_E}{\partial \xi_i} |_{(\xi_{1,E}^g, \xi_{2,E}^g)}, \quad i = 1, 2,$$

$$|\mathbf{g}_3| = |\mathbf{g}_1 \times \mathbf{g}_2|.$$

 For $A = 1, \dots, n_{\text{ino}}$ do:

 For $B = 1, \dots, n_{\text{ino}}$ do:

$$\mathbf{A}_E^{AB} \leftarrow \mathbf{A}_E^{AB} + \beta \hat{N}^A \hat{N}^B |\mathbf{g}_3| w_E^g$$

 End do

$$\mathbf{f}_E^A \leftarrow \mathbf{f}_E^A + \hat{N}^A g(\mathbf{x}_E^g) |\mathbf{g}_3| w_E^g$$

 End do

End do

End do

Note that not all the nodes of element K will contribute to the integral over E , as their shape functions there can be zero. These shape functions, as well as vectors $\mathbf{g}_i$, $i = 1, 2, 3$, are all evaluated at $(\xi_{1,E}^g, \xi_{2,E}^g)$ in the previous algorithm, although we have not specified this point to lighten the notation.

1.3.4 Basic estimates

Interpolation estimates

The FE method described yields a solution $u_h \in V_h$. As it has been said, the objective of the FE analysis is to express the error $u - u_h$ in terms of the interpolation error given in Eq. (1.20). Once we know how FE spaces are constructed, let us give some basic estimates for this interpolation error.

We will start with a very simple estimate for P_1 -elements that is only valid for smooth functions, but that has the main features of interpolation estimates. In particular, it displays how this estimate degrades when the FE partition is not shape regular. For its instructiveness, we shall give the proof, which uses basic mathematical tools. Then we will present another estimate for P_1 -elements, now on shape regular partitions but for more general functions. In both cases the solution to be interpolated will be continuous, so that we can define its nodal interpolation:

$$\pi_h(v)(\mathbf{x}) = \sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x}) v^a, \quad v^a = v(\mathbf{x}^a). \quad (1.36) \quad \{\text{eq:c1-nod}\}$$

In the case of the interpolation of functions v that are not continuous, the nodal interpolation is not well defined and other alternatives need to be devised. In general, one considers mappings from $L^1(\omega_K)$ to an appropriate space of polynomials on ω_K that allow one to define nodal values in an average sense, where ω_K is a domain that contains K . With these nodal values the standard FE interpolation can be used. Examples of these mappings, called **FE interpolants**, are Clément's and Scott-Zhang's interpolants. The former is perhaps the most intuitive polynomial average interpolator, whereas the latter is a modified one designed to preserve boundary values. There are, however, many variants. Obviously, FE interpolation theory is outside the scope of these notes.

All results will be first presented within each element domain $K \in \mathcal{T}_h$. The extension to estimates for the whole domain Ω will be straightforward in the case of quasi-uniform FE partitions.

From now on, we will use the symbol $\lesssim$ to denote $\leq$ up to positive *dimensionless* constants, and likewise for $\gtrsim$.

Theorem 3 Suppose that $v \in W^{2,\infty}(K)$. Then, for P_1 elements there holds:

$$\|v - \pi_h(v)\|_{L^\infty(K)} \lesssim h_K^2 \max_{|\alpha|=2} \|D^\alpha v\|_{L^\infty(K)}, \quad (1.37)$$

$$\max_{|\alpha|=1} \|D^\alpha(v - \pi_h(v))\|_{L^\infty(K)} \lesssim \frac{h_K^2}{\rho_K} \max_{|\alpha|=2} \|D^\alpha(v - \pi_h(v))\|_{L^\infty(K)}. \quad (1.38)$$

Proof. Since functions v that belong to $W^{2,\infty}(K)$ are continuous, we can consider their nodal interpolation given by Eq. (1.36). A Taylor expansion around $\mathbf{x} \in K$ allows us to write the nodal value v^a as

$$v^a = v(\mathbf{x}^a) = v(\mathbf{x}) + P^a(\mathbf{x}) + R^a(\mathbf{x}),$$

$$P^a(\mathbf{x}) := \frac{\partial v}{\partial x_k}(\mathbf{x})(x_k^a - x_k), \quad R^a(\mathbf{x}) := \frac{1}{2} \frac{\partial^2 v}{\partial x_k \partial x_l}(\bar{\mathbf{x}})(x_k^a - x_k)(x_l^a - x_l),$$

for a certain $\bar{\mathbf{x}} \in K$ (recall that repeated indexes imply summation over d). Therefore:

$$\pi_h(v)(\mathbf{x}) = \sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x})[v(\mathbf{x}) + P^a(\mathbf{x}) + R^a(\mathbf{x})].$$

Using that the sum of shape functions is 1 and the concept of isoparametric mapping:

$$\sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x})v(\mathbf{x}) = v(\mathbf{x}) \sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x}) = v(\mathbf{x})$$

$$\sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x})P^a(\mathbf{x}) = \frac{\partial v}{\partial x_k}(\mathbf{x}) \left[\sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x})x_k^a - x_k \sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x}) \right] = 0,$$

from where:

$$\pi_h(v)(\mathbf{x}) - v(\mathbf{x}) = \sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x})R^a(\mathbf{x}).$$

Estimate (1.37) follows noticing that

$$|R^a(\mathbf{x})| \lesssim h_K^2 \max_{|\alpha|=2} \|D^\alpha v\|_{L^\infty(K)}.$$

Note that the constant involved is d .

Let us prove now (1.38). We have that

$$\frac{\partial \pi_h(v)}{\partial x_m}(\mathbf{x}) = \sum_{a=1}^{n_{\text{ino}}} \frac{\partial N^a}{\partial x_m}(\mathbf{x}) v^a = \sum_{a=1}^{n_{\text{ino}}} \frac{\partial N^a}{\partial x_m}(\mathbf{x}) [v(\mathbf{x}) + P^a(\mathbf{x}) + R^a(\mathbf{x})].$$

The first term is zero, as the sum of shape functions derivatives is zero. For the second term we have:

$$\sum_{a=1}^{n_{\text{ino}}} \frac{\partial N^a}{\partial x_m}(\mathbf{x}) P^a(\mathbf{x}) = \frac{\partial v}{\partial x_k}(\mathbf{x}) \frac{\partial}{\partial x_m} \sum_{a=1}^{n_{\text{ino}}} N^a(\mathbf{x}) x_k^a = \frac{\partial v}{\partial x_k}(\mathbf{x}) \frac{\partial x_k}{\partial x_m} = \frac{\partial v}{\partial x_m}(\mathbf{x}).$$

Therefore:

$$\frac{\partial \pi_h(v)}{\partial x_m}(\mathbf{x}) - \frac{\partial v}{\partial x_m}(\mathbf{x}) = \sum_{a=1}^{n_{\text{ino}}} \frac{\partial N^a}{\partial x_m}(\mathbf{x}) R^a(\mathbf{x}).$$

Estimate (1.38) follows noticing that $\frac{\partial N^a}{\partial x_m}(\mathbf{x}) \leq \rho_K^{-1}$. The constant involved in this case is dn_{ino} . $\square$

Theorem 4 Suppose that the FE partition is shape regular and let $v \in H^2(K)$. Then, for P_1 elements there holds:

$$|v - \pi_h(v)|_{H^1(K)} \lesssim h_K |v|_{H^2(K)}.$$

Proof. We will use a particular case of the Sobolev embedding theorem: given a bounded domain $\omega \subset \mathbb{R}^d$, if $mp > d$ then $W^{m,p}(\omega) \hookrightarrow C^0(\bar{\omega})$ and for any $v \in W^{m,p}(\omega)$ there holds $\|v\|_{L^\infty(\omega)} \lesssim \ell_\omega^{d/p} \|v\|_{W^{m,p}(\omega)}$ (recall that we have introduced the length scale ℓ_ω in the definition of $\|\cdot\|_{W^{m,p}(\omega)}$).

As functions in $H^2(K)$ happen to be continuous on $\bar{K}$, the nodal interpolation is again well defined. Let $\hat{v} = v \circ \varphi_K$. In this case, φ_K is affine, and

$$\pi_h(\hat{v}) = \pi_h(v) \circ \varphi_K,$$

where the nodal interpolation in $\hat{K}$ is still denoted π_h . This is easily proved checking that both sides of the equation are polynomials of degree one that coincide at the nodes of $\hat{K}$.

Let $\hat{p} \in P_1(\hat{K})$. Obviously, $\pi_h(\hat{p}) = \hat{p}$, and in the reference element we have:

$$|\hat{v} - \pi_h(\hat{v})|_{H^1(\hat{K})} = |(\hat{v} + \hat{p}) - \pi_h(\hat{v} + \hat{p})|_{H^1(\hat{K})} \leq |\hat{v} + \hat{p}|_{H^1(\hat{K})} + |\pi_h(\hat{v} + \hat{p})|_{H^1(\hat{K})}.$$

From the Sobolev embedding theorem stated, since $1 \leq \ell_{\hat{K}} \leq 2$:

$$|\pi_h(\hat{v} + \hat{p})|_{H^1(\hat{K})} \lesssim \|\pi_h(\hat{v} + \hat{p})\|_{L^\infty(\hat{K})} \lesssim \|\hat{v} + \hat{p}\|_{L^\infty(\hat{K})} \lesssim \|\hat{v} + \hat{p}\|_{H^2(\hat{K})},$$

and therefore

$$|\hat{v} - \pi_h(\hat{v})|_{H^1(\hat{K})} \lesssim \|\hat{v} + \hat{p}\|_{H^2(\hat{K})}, \quad \forall \hat{p} \in P_1(\hat{K}).$$

The so called Bramble-Hilbert lemma guarantees that

$$\inf_{\hat{p} \in P_1(\hat{K})} \|\hat{v} + \hat{p}\|_{H^2(\hat{K})} \lesssim |\hat{v}|_{H^2(\hat{K})},$$

from where we obtain

$$|\hat{v} - \pi_h(\hat{v})|_{H^1(\hat{K})} \lesssim |\hat{v}|_{H^2(\hat{K})}.$$

For a shape regular partition, a simple change of variables in the integral (often called scaling argument) yields

$$h_K^{m-d/p} |w|_{W^{m,p}(K)} \lesssim |w \circ \varphi_K|_{W^{m,p}(\hat{K})} \lesssim h_K^{m-d/p} |w|_{W^{m,p}(K)},$$

for any $w \in W^{m,p}(K)$. Using this with $p = 2$ and $m = 1, 2$ we get

$$h_K^{1-d/2} |v - \pi_h(v)|_{H^1(K)} \lesssim h_K^{2-d/2} |v|_{H^2(K)},$$

from where the theorem follows. $\square$

The following result generalises this theorem to more general Sobolev spaces, arbitrary interpolations and functions that are not necessarily continuous. It also displays how the interpolation estimates deteriorate with the aspect ratio of the element subdomains:

Theorem 5 *Let $v \in W^{r,p}(K)$. Then, there exists an interpolant $\tilde{v}_h \in R_k(K)$ such that*

$$|v - \tilde{v}_h|_{W^{m,p}(K)} \lesssim \frac{h_K^\mu}{\rho_K^m} |v|_{W^{\mu,p}(K)}, \quad \mu := \min(r, k+1).$$

So far we have considered the increase in the dimension of V_h by increasing the number of subdomains, i.e., reducing h ; as mentioned earlier, this is called h -refinement. However, we could also consider increasing the polynomial order k to improve the FE approximation; this is called k -**refinement**. In fact, the polynomial order is mostly called p , in the literature, and the process of increasing this order is called p -refinement. However, these notes are intended to approximate problems in fluid mechanics, and p will be used exclusively for the pressure. The strategy of both decreasing h and increasing k is called hk -refinement, and the FE space in this case is often represented as V_{hk} . The following result is a basic interpolation estimate for hk -refinement:

Theorem 6 *Let $v \in H^r(K)$ and suppose that the FE partition is shape regular. Then, there exists an interpolant $\tilde{v}_h \in R_k(K)$ such that*

$$|v - \tilde{v}_h|_{H^m(K)} \lesssim \frac{h_K^{\mu-m}}{k^{\mu-m}} |v|_{H^\mu(K)}, \quad \mu := \min(r, k+1). \quad (1.39)$$

Suppose that $r > k+1$ and take for example $m = 1$. We then have:

$$|v - \tilde{v}_h|_{H^1(K)} \lesssim \frac{h_K^k}{k^k} |v|_{H^{k+1}(K)}.$$

It is observed that when k is increased the interpolation error decreases as k^k (apart from the factor h_K^k). This is referred to as **exponential convergence**. Note however that it requires the interpolated solution to be smooth enough, i.e., with Sobolev regularity $r \geq k+1$. In h -refinement, the regularity required for the function being interpolated is fixed.

Finally, let us check how to extend the local interpolation estimates provided to the whole domain, for example with estimate (1.39) and assuming $r \geq k+1$. We have that:

$$\|v - \tilde{v}_h\|_{H^m(\Omega)}^2 = \sum_{n=0}^m \sum_{|\alpha|=n} \sum_K \ell_\Omega^{2n} \|D^\alpha v - D^\alpha \tilde{v}_h\|_{L^2(K)}^2 \lesssim \sum_{n=0}^m \sum_K \ell_\Omega^{2n} \frac{h_K^{2(k+1-n)}}{k^{2(k+1-n)}} |v|_{H^{k+1}(K)}^2.$$

If the mesh is quasi-uniform:

$$\|v - \tilde{v}_h\|_{H^m(\Omega)}^2 \lesssim \sum_{n=0}^m \frac{h^{2(k+1-n)}}{k^{2(k+1-n)}} \ell_\Omega^{2n} \sum_K |v|_{H^{k+1}(K)}^2 = \sum_{n=0}^m \frac{h^{2(k+1-n)}}{k^{2(k+1-n)}} \ell_\Omega^{2n} |v|_{H^{k+1}(\Omega)}^2,$$

from where

$$\|v - \tilde{v}_h\|_{H^m(\Omega)} \lesssim \frac{h^{k+1}}{k^{k+1}} |v|_{H^{k+1}(\Omega)} \sum_{n=0}^m \frac{\ell_\Omega^n k^n}{h^n} \lesssim \ell_\Omega^m \frac{h^{k+1-m}}{k^{k+1-m}} |v|_{H^{k+1}(\Omega)}. \quad (1.40) \quad \{\text{eq:c1-bas}\}$$

All other estimates can be treated analogously.

Inverse estimates and trace inequalities

Let us conclude this section stating without proof two inequalities that will be used in these notes. The first establishes that given a polynomial function v_h of order k in K there exists a constant C_{inv} such that:

$$\|\nabla v_h\|_{L^2(K)} \leq C_{\text{inv}} \frac{k^2}{h_K} \|v_h\|_{L^2(K)}. \quad (1.41) \quad \{\text{eq:c1esti}\}$$

This inequality can be used for scalars, vectors or tensors. It is called **inverse inequality**, as it bounds a norm of the derivative of a function with a norm of the function itself. Clearly, it has to explode if $k \rightarrow \infty$ or if $h_K \rightarrow 0$, as it cannot hold for general functions. The factor h_K^{-1} is obvious using a scaling argument, but it is important to remark the factor k^2 , that is known to be sharp. Let us also observe that since derivatives of polynomials are polynomials, it can be applied to any derivative.

Similarly, the **trace inequality**

$$\|v\|_{L^2(\partial K)}^2 \leq C_{\text{trace}} \left(h_K^{-1} \|v\|_{L^2(K)}^2 + h_K \|\nabla v\|_{L^2(K)}^2 \right),$$

can be shown to hold for functions $v \in H^1(K)$, $K \in \mathcal{T}_h$, for pretty general FE partitions and for a constant C_{trace} . The important point is that the last term can be dropped if v is a polynomial on the element domain K . If v_h is a polynomial of degree k , there holds

$$\|v_h\|_{L^2(\partial K)} \leq C_{\text{trace}} \frac{k}{h_K^{1/2}} \|v_h\|_{L^2(K)}. \quad (1.42) \quad \{\text{eq:c1-tra}\}$$

We have still denoted C_{trace} the constant involved in this inequality. Again, it has to explode if $k \rightarrow \infty$ or if $h_K \rightarrow 0$, as functions in $L^2(K)$ do not necessarily belong to $L^2(\partial K)$. The factor $h_K^{-1/2}$ can be justified by a simple scaling, and again the factor k needs to be highlighted.

Summary

Let us collect the most important estimates we have introduced:

$ v - \tilde{v}_h _{H^m(K)} \lesssim \frac{h_K^{k+1-m}}{k^{k+1-m}} v _{H^{k+1}(K)},$	Local interpolation error estimate, shape regular $\mathcal{T}_h$, $v \in H^{k+1}(K)$.
$\ v - \tilde{v}_h\ _{H^m(\Omega)} \lesssim \ell_\Omega^m \frac{h_K^{k+1-m}}{k^{k+1-m}} v _{H^{k+1}(\Omega)}.$	Global interpolation error estimate, quasi-uniform $\mathcal{T}_h$, $v \in H^{k+1}(\Omega)$.
$\ \nabla v_h\ _{L^2(K)} \leq C_{\text{inv}} \frac{k^2}{h_K} \ v_h\ _{L^2(K)}$	Inverse inequality.
$\ v_h\ _{L^2(\partial K)} \leq C_{\text{trace}} \frac{k}{h_K^{1/2}} \ v_h\ _{L^2(K)}$	Trace inequality.

It is important to remark that all these inequalities are asymptotic, i.e., they hold for h_K small enough and k large enough.

1.3.5 Duality arguments

From Céa's lemma (1.19) and the definition (1.20) we may conclude that

$$\|u - u_h\|_V \leq \frac{N_a}{K_a} E_{\text{int}}(h, u) \leq \frac{N_a}{K_a} \inf_{v_h \in V_h} \|u - v_h\|_V.$$

In the case of Poisson's problem, the norm of V is the $H^1(\Omega)$ norm. Using the interpolation estimate (1.40) with $m = 1$, for the solution of the continuous problem u regular enough, $\mathcal{T}_h$ quasi-uniform and using elements of degree k :

$$\|u - u_h\|_{H^1(\Omega)} \lesssim \frac{N_a}{K_a} \ell_\Omega \frac{h_K^k}{k^k} |u|_{H^1(\Omega)}.$$

The question that now arises is whether it is possible to obtain an error estimate in the $L^2(\Omega)$ norm. This is indeed possible, but we need an additional condition to hold. Let us write explicitly the bound (1.16):

$$\|u\|_{H^1(\Omega)} \lesssim \frac{1}{K_a} \left(\|f\|_{H^{-1}(\Omega)} + \|g\|_{H^{-1/2}(\Gamma_g)} \right).$$

The question is now: is it possible to guarantee higher Sobolev regularity for u if the regularity of f and g is increased? The answer is positive under certain conditions on the domain Ω . Suppose that $\Gamma_g = \emptyset$. It can be shown for example that if $d = 2$ and Ω is polyhedral or with a C^2 boundary, there holds

$$\|u\|_{H^2(\Omega)} \lesssim C_{\text{reg}} \frac{1}{K_a} \|f\|, \tag{1.43}$$

where C_{reg} is a constant that depends on Ω . This is known as an **elliptic regularity** property or **shift condition**. Whether conditions of this type hold or not is a delicate question in the analysis of continuous partial differential equations. In numerical analysis, we need to assume they hold to get improved error estimates, as in the following result. The technique employed in the proof is called **Aubin-Nitsche's duality argument**.

Theorem 7 Consider the problem of finding $u_h \in V_h$ such that $a(u_h, v_h) = (f, v_h)$ for all $v_h \in V_h$, with a continuous and coercive, and $f \in L^2(\Omega)$. Suppose also that the elliptic regularity condition (1.43) holds. Then:

$$\|u - u_h\| \lesssim h \|u - u_h\|_{H^1(\Omega)}. \tag{1.44}$$

Proof. Let us consider the following continuous problem: find $w \in V$ such that

$$a(v, w) = (u - u_h, v) \quad \forall v \in V.$$

Observe that the test function is in the first slot of a , although in our case this is irrelevant, since a is symmetric. In general, this corresponds to solving for the adjoint of the original differential operator. Because of condition (1.43) the solution to this problem will satisfy

$$\|w\|_{H^2(\Omega)} \lesssim C_{\text{reg}} \frac{1}{K_a} \|u - u_h\|, \quad (1.45) \quad \{\text{eq:c1-e11}\}$$

Let $\tilde{w}_h \in V_h$ be an optimal interpolant for w . Taking now $v = u - u_h$ we get:

$$\begin{aligned} \|u - u_h\|^2 &= a(u - u_h, w) \\ &= a(u - u_h, w - \tilde{w}_h) && \text{Galerkin orthogonality,} \\ &\leq N_a \|u - u_h\|_{H^1(\Omega)} \|w - \tilde{w}_h\|_{H^1(\Omega)} && \text{Continuity of } a, \\ &\lesssim N_a \|u - u_h\|_{H^1(\Omega)} h \|w\|_{H^2(\Omega)} && \text{Interpolation error for } w, \\ &\lesssim N_a \|u - u_h\|_{H^1(\Omega)} h C_{\text{reg}} \frac{1}{K_a} \|u - u_h\| && \text{Elliptic regularity,} \end{aligned}$$

from where the result follows. $\square$

This type of duality techniques can be applied to more general problems, although here we have considered only the simplest setting for Poisson's problem. In the case of regular enough solutions u , estimate (1.44) states that if $\|u - u_h\|_{H^1(\Omega)}$ is $\mathcal{O}(h^k)$, then $\|u - u_h\|$ is $\mathcal{O}(h^{k+1})$, i.e., of the order of the interpolation error in $L^2(\Omega)$. However, in the proof it is observed that the (dimensionless) constant involved in this estimate behaves as $C_{\text{reg}} N_a / K_a$, and thus it may deteriorate depending on the shape of Ω and the ratio N_a / K_a .

1.4 Mixed form of Poisson's problem

1.4.1 Boundary value problem

Up to this point we have considered the so called **irreducible** form of Poisson's, given by Eq. (1.1) and the boundary conditions (1.2)-(1.3). However, very often it is convenient to write the problem introducing as a new variable the flux of u , i.e., $\mathbf{q} = -\kappa \nabla u$. This is the case for example when u is the temperature, or the pressure in a porous medium. The reason for introducing $\mathbf{q}$ as a variable may be that we want a better approximation for it, as we shall see. Since this introduction will imply dealing with two different variables, $\mathbf{q}$ and u , the resulting formulation will be called **mixed**.

The mixed form of Poisson's problem we shall consider is the following: find $\mathbf{q} : \Omega \rightarrow \mathbb{R}^d$ and $u : \Omega \rightarrow \mathbb{R}$ such that

$$\kappa^{-1} \mathbf{q} + \nabla u = 0 \quad \text{in } \Omega, \quad (1.46) \quad \{\text{eq:c1-d1}\}$$

$$\nabla \cdot \mathbf{q} + \sigma u = f \quad \text{in } \Omega, \quad (1.47) \quad \{\text{eq:c1-d2}\}$$

$$-\beta_u \mathbf{n} \cdot \mathbf{q} + u = \bar{u} \quad \text{on } \Gamma_u, \quad (1.48) \quad \{\text{eq:c1-d3}\}$$

$$-\mathbf{n} \cdot \mathbf{q} + \beta_g u = g \quad \text{on } \Gamma_g. \quad (1.49) \quad \{\text{eq:c1-d4}\}$$

This is a generalised version of **Darcy's problem**, which models the flow in a porous medium and therefore can be considered as the first fluid mechanics model we introduce. In this case, $\sigma = 0$, κ ,

which may be a tensor, is called **permeability**, u is a pressure and $\mathbf{q}$ a velocity. In general, u is called the **primal variable** and $\mathbf{q}$ the **dual variable**.

The way we have written the boundary conditions needs an explanation. Obviously, Eq. (1.49) is the same as Eq. (1.3) introducing $\mathbf{q}$ (and replacing β by β_g). Concerning the Robin condition (1.48), it tends to $u = \bar{u}$ when $\beta_u \rightarrow 0$. However, we shall see that in one of the functional settings of the problem described next the boundary conditions on u will be enforced weakly, whereas that for $\mathbf{n} \cdot \mathbf{q}$ will be enforced strongly. This is why in this case it makes sense to prescribe a Robin type condition on Γ_u .

1.4.2 Primal mixed form

Let us start considering the case in which the functional setting is the same as for the irreducible problem studied earlier. Since in that case we had $u \in H^1(\Omega)$ and now $\mathbf{q} = -\kappa \nabla u$, we will have that $\mathbf{q} \in L^2(\Omega)^d$. Likewise, we will consider $\beta_u = 0$ and $\bar{u} = 0$ in (1.48) and prescribe this condition strongly, whereas condition (1.49) will be prescribed weakly.

Let $V = \{v \in H^1(\Omega) \mid v = 0 \text{ on } \Gamma_u\}$ and $R = L^2(\Omega)^d$. Consider arbitrary test functions $v \in V$ and $\mathbf{r} \in R$. Multiplying Eq. (1.46) by $\mathbf{r}$ and Eq. (1.47) by v and integrating over Ω we get:

$$\kappa^{-1} \int_{\Omega} \mathbf{q} \cdot \mathbf{r} + \int_{\Omega} \nabla u \cdot \mathbf{r} = 0, \quad (1.50)$$

$$\int_{\Omega} v \nabla \cdot \mathbf{q} + \sigma \int_{\Omega} uv = \int_{\Omega} vf. \quad (1.51)$$

The first term in (1.51) is not well defined if $\mathbf{q}$ belongs only to $L^2(\Omega)^d$. If we integrate it by parts we get

$$\int_{\Omega} v \nabla \cdot \mathbf{q} = - \int_{\Omega} \nabla v \cdot \mathbf{q} + \int_{\partial\Omega} v \mathbf{n} \cdot \mathbf{q} = - \int_{\Omega} \nabla v \cdot \mathbf{q} + \beta_g \int_{\Gamma_g} vu - \int_{\Gamma_g} vg.$$

Using this together with Eqs. (1.50)-(1.51) we get the following problem: find $u \in V$ and $\mathbf{q} \in R$ such that

$$\kappa^{-1}(\mathbf{q}, \mathbf{r}) + (\nabla u, \mathbf{r}) = 0 \quad \forall \mathbf{r} \in R, \quad (1.52)$$

$$-(\nabla v, \mathbf{q}) + \sigma(v, u) + \beta_g(v, u)_{L^2(\Gamma_g)} = \langle v, f \rangle + \langle v, g \rangle_{\Gamma_g} \quad \forall v \in V. \quad (1.53)$$

It is observed that all terms are well defined. The problem obtained is analogous to that of the irreducible form, now having introduced $\mathbf{q}$ which, because of Eq. (1.52), is equal to $\kappa \nabla u$ in $L^2(\Omega)^d$. This is the so called **primal mixed form** of the problem. In the case of constant κ it has no practical interest at all, but in the case of variable κ (not treated here) the flux $\mathbf{q}$ can be understood as the projection of $\kappa \nabla u$ onto $L^2(\Omega)^d$.

Let us define the bilinear forms:

$$a : R \times R \longrightarrow \mathbb{R} \\ \mathbf{q}, \mathbf{r} \mapsto a(\mathbf{q}, \mathbf{r}) := \kappa^{-1}(\mathbf{q}, \mathbf{r}), \quad (1.54)$$

$$b : V \times R \longrightarrow \mathbb{R} \\ v, \mathbf{r} \mapsto b(v, \mathbf{r}) := (\nabla v, \mathbf{r}), \quad (1.55)$$

$$c : V \times V \longrightarrow \mathbb{R} \\ v, u \mapsto c(u, v) := \sigma(v, u) + \beta_g(v, u)_{L^2(\Gamma_g)}, \quad (1.56)$$

and the linear form:

$$\begin{aligned} l_u : V &\longrightarrow \mathbb{R} \\ v &\mapsto l_u(v) := \langle v, f \rangle + \langle v, g \rangle_{\Gamma_g}. \end{aligned} \quad (1.57)$$

It is also convenient to introduce $l_q \in R'$, now defined by $l_q(\mathbf{r}) = 0$. Problem (1.52)-(1.53) can now be written as follows: find $u \in V$ and $\mathbf{q} \in R$ such that

$$a(\mathbf{q}, \mathbf{r}) + b(u, \mathbf{r}) = l_q(\mathbf{r}), \quad \forall \mathbf{r} \in R, \quad (1.58) \quad \{\text{eq:c1-d1b}\}$$

$$-b(v, \mathbf{q}) + c(u, v) = l_u(v), \quad \forall v \in V. \quad (1.59) \quad \{\text{eq:c1-d2b}\}$$

This system can be written as a single variational equation as follows: find $U := [\mathbf{q}, u] \in R \times V =: X$ such that

$$B(U, V) = L(V) \quad \forall V := [\mathbf{r}, v] \in X \quad (1.60) \quad \{\text{eq:c1-mix}\}$$

where

$$\begin{aligned} B(U, V) &:= a(\mathbf{q}, \mathbf{r}) + b(u, \mathbf{r}) - b(v, \mathbf{q}) + c(u, v), \\ L(V) &:= l_q(\mathbf{r}) + l_u(v). \end{aligned}$$

We will use capital letters to denote the forms, unknowns and test functions of the mixed problem written as a single equation. Note that the symbol V may be either the space for u or the test function in the product space X . The meaning will be clear by the context; we prefer this possible confusion than overloading the notation.

If we introduce the linear operators:

$$\begin{aligned} A_a : R &\longrightarrow R', & A_a \mathbf{q} &:= a(\mathbf{q}, \cdot), \\ A_b : V &\longrightarrow R', & A_b u &:= b(u, \cdot), \\ A_b^T : R &\longrightarrow V', & A_b^T \mathbf{q} &:= b(\cdot, \mathbf{q}), \\ A_c : V &\longrightarrow V', & A_c u &:= c(u, \cdot), \\ f_q &\in R', & f_q &= l_q(\cdot), \\ f_u &\in V', & f_u &= l_u(\cdot), \end{aligned}$$

problem (1.58)-(1.59) can be written as:

$$\begin{bmatrix} A_a & A_b \\ -A_b^T & A_c \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ u \end{bmatrix} = \begin{bmatrix} f_q \\ f_u \end{bmatrix}. \quad (1.61) \quad \{\text{eq:c1-dop}\}$$

This formalism has a direct counterpart in the matrix form of the problem when it is discretised. Observe that in the way we have written the problem the bilinear form B is positive, i.e., $B(U, U) = a(\mathbf{q}, \mathbf{q}) + c(u, u) > 0$ for $U \neq 0$, but it is not symmetric, since $B(U, V) - B(V, U) = 2b(u, \mathbf{r}) - 2b(v, \mathbf{q}) \neq 0$ in general. If the sign of Eq. (1.59) is changed, then the resulting form would be symmetric, but not definite. These facts obviously translate to the previous operator equation.

1.4.3 Dual mixed form

The previous mixed form is unnecessary, as it is equivalent (at the continuous level) to the irreducible form. However, we can change the function setting and still have a well posed problem. We will follow

the same formal steps as before, but with *a change in the term integrated by parts*. Starting from system (1.50)-(1.51) now we integrate by parts the following term in Eq. (1.50):

$$\int_{\Omega} \nabla u \cdot \mathbf{r} = - \int_{\Omega} u \nabla \cdot \mathbf{r} + \int_{\partial\Omega} u \mathbf{n} \cdot \mathbf{r},$$

and therefore, before prescribing boundary conditions, the equations obtained by testing the original differential equations by $\mathbf{r}$ and v are:

$$\begin{aligned} \kappa^{-1} \int_{\Omega} \mathbf{q} \cdot \mathbf{r} - \int_{\Omega} u \nabla \cdot \mathbf{r} + \int_{\partial\Omega} u \mathbf{n} \cdot \mathbf{r} &= 0, \\ \int_{\Omega} v \nabla \cdot \mathbf{q} + \sigma \int_{\Omega} uv &= \int_{\Omega} vf. \end{aligned}$$

Now there are no derivatives of u and v , but we have the divergence of $\mathbf{q}$ and $\mathbf{r}$. It is clear that the volume integrals are well defined if $u, v \in L^2(\Omega)$ and $\nabla \cdot \mathbf{q}, \nabla \cdot \mathbf{r} \in L^2(\Omega)$. This leads to introduce the space:

$$\{eq:c1-hdiv\} \quad H(\text{div}, \Omega) := \{\mathbf{r} : \Omega \longrightarrow \mathbb{R}^d \mid \mathbf{r} \in L^2(\Omega)^d \text{ and } \nabla \cdot \mathbf{r} \in L^2(\Omega)\}, \quad (1.62)$$

which turns out to be a Hilbert space when endowed with the inner product

$$(\mathbf{q}, \mathbf{r})_{H(\text{div}, \Omega)} := (\mathbf{q}, \mathbf{r}) + \ell_{\Omega}^2(\nabla \cdot \mathbf{q}, \nabla \cdot \mathbf{r}),$$

and associated norm

$$\|\mathbf{r}\|_{H(\text{div}, \Omega)} := \|\mathbf{r}\| + \ell_{\Omega} \|\nabla \cdot \mathbf{r}\|.$$

It can be proved that functions $\mathbf{r} \in H(\text{div}, \Omega)$ have a well defined normal trace $\mathbf{n} \cdot \mathbf{r} \in H^{-1/2}(\partial\Omega)$, the trace operator being continuous (bounded). Note in particular that the duality $H^{-1/2}(\partial\Omega) \times H^{1/2}(\partial\Omega)$ will be

$$\langle \mathbf{n} \cdot \mathbf{r}, v \rangle_{H^{-1/2}(\partial\Omega) \times H^{1/2}(\partial\Omega)} = \int_{\partial\Omega} \mathbf{n} \cdot \mathbf{r} v = \int_{\Omega} (\nabla \cdot \mathbf{r} v + \mathbf{r} \cdot \nabla v),$$

all terms being well defined for $v \in H^1(\Omega)$, $v|_{\partial\Omega} \in H^{1/2}(\partial\Omega)$, $\mathbf{r} \in H(\text{div}, \Omega)$.

As boundary conditions we will now consider (1.49) with $\beta_g = 0$, $g = 0$, *now imposed strongly*, and (1.48) *imposed weakly*. We thus define the spaces:

$$\begin{aligned} R &= \{\mathbf{r} \in H(\text{div}, \Omega) \mid \mathbf{n} \cdot \mathbf{r} = 0 \text{ on } \Gamma_g \text{ and } \mathbf{n} \cdot \mathbf{r} \in L^2(\Gamma_u) \text{ if } \beta_u \neq 0\}, \\ V &= L^2(\Omega). \end{aligned}$$

The problem to be solved is then the following: find $u \in V$ and $\mathbf{q} \in R$ such that

$$\{eq:c1-d1d1\} \quad \kappa^{-1}(\mathbf{q}, \mathbf{r}) + \beta_u(\mathbf{n} \cdot \mathbf{q}, \mathbf{n} \cdot \mathbf{r})_{\Gamma_u} - (u, \nabla \cdot \mathbf{r}) = -\langle \bar{u}, \mathbf{n} \cdot \mathbf{r} \rangle_{\Gamma_u} \quad \forall \mathbf{r} \in R, \quad (1.63)$$

$$\{eq:c1-d1d2\} \quad (v, \nabla \cdot \mathbf{q}) + \sigma(v, u) = (v, f) \quad \forall v \in V. \quad (1.64)$$

Let us comment on the regularity of the data. First, the reason for taking $g = 0$ in (1.49) is simply to have the same space for test functions and trial solutions for $\mathbf{q}$. Since $v \in L^2(\Omega)$, now we need also $f \in L^2(\Omega)$. If $\beta_u = 0$, since $\mathbf{n} \cdot \mathbf{r} \in H^{-1/2}(\Gamma_g)$, the term $\langle \bar{u}, \mathbf{n} \cdot \mathbf{r} \rangle_{\Gamma_u}$ would require $\bar{u} \in H^{1/2}(\Gamma_g)$. However, for $\beta_u > 0$ requiring the term $(\mathbf{n} \cdot \mathbf{q}, \mathbf{n} \cdot \mathbf{r})_{\Gamma_u}$ to be bounded leads to ask for $\mathbf{n} \cdot \mathbf{r} \in L^2(\Gamma_g)$ for all elements in R , and in this case we only need $\bar{u} \in L^2(\Gamma_g)$.

The functional setting of problem (1.63)-(1.64) is different to that of the irreducible problem. It is the **dual mixed form** of Poisson's problem, or the dual form of the generalised Darcy problem. In brief, the regularity requirements for u have been relaxed at the expense of increasing the regularity of $\mathbf{q}$. This has important consequences in the numerical approximation, as now it is possible to achieve better accuracy for $\mathbf{q}$ than in the primal form, and this is interesting in many applications in which $\mathbf{q}$ is the variable of interest. This is usually the case in the flow of porous media.

We can now introduce the bilinear forms:

$$a : R \times R \longrightarrow \mathbb{R} \\ \mathbf{q}, \mathbf{r} \mapsto a(\mathbf{q}, \mathbf{r}) := \kappa^{-1}(\mathbf{q}, \mathbf{r}) + \beta_u(\mathbf{n} \cdot \mathbf{q}, \mathbf{n} \cdot \mathbf{r})_{\Gamma_u}, \quad (1.65)$$

$$b : V \times R \longrightarrow \mathbb{R} \\ v, \mathbf{r} \mapsto b(v, \mathbf{r}) := -(v, \nabla \cdot \mathbf{r}), \quad (1.66)$$

$$c : V \times V \longrightarrow \mathbb{R} \\ v, u \mapsto c(u, v) := \sigma(v, u), \quad (1.67)$$

and the linear forms:

$$l_q : R \longrightarrow \mathbb{R} \\ v \mapsto l_q(\mathbf{r}) := -\langle \bar{u}, \mathbf{n} \cdot \mathbf{r} \rangle_{\Gamma_u}, \quad (1.68)$$

$$l_u : V \longrightarrow \mathbb{R} \\ v \mapsto l_u(v) := (v, f). \quad (1.69)$$

This allows us to write the mixed dual form of the problem formally as the mixed primal one. In particular, the problem can be written as indicated in Eqs. (1.58), (1.59) and (1.60), with the new definition of spaces R , V and the forms in play. The operator form of the problem is again given by Eq. (1.61). The structure of this equation is found in other problems: the diagonal operators are positive semi-definite and the off-diagonal operators are one equal to minus the transpose of the other.

1.4.4 Basic theory of inf-sup stable problems

The problem to be solved in the mixed form of Poisson's problem is (1.60). The difficulty is that the bilinear form B is continuous, but *not coercive*. To see this, let us consider the following norm on $X = R \times V$:

$$\|[\mathbf{r}, u]\|_X^2 := \kappa^{-1}\|\mathbf{r}\|^2 + \text{if}_d \kappa^{-1} \ell_\Omega^2 \|\nabla \cdot \mathbf{r}\|^2 + \kappa \ell_\Omega^{-2} \|v\|^2 + (1 - \text{if}_d) \kappa \|\nabla v\|^2,$$

where $\text{if}_d = 1$ for the dual form and $\text{if}_d = 0$ for the primal form. It is readily checked that

$$B(U, U) = a(\mathbf{q}, \mathbf{q}) + c(u, u) = \begin{cases} \kappa^{-1}\|\mathbf{q}\|^2 + \sigma\|u\|^2 + \beta_g\|u\|_{L^2(\Gamma_g)} & \text{for the primal form,} \\ \kappa^{-1}\|\mathbf{q}\|^2 + \beta_u\|\mathbf{n} \cdot \mathbf{q}\|_{L^2(\Gamma_g)}^2 + \sigma\|u\|^2 & \text{for the dual form.} \end{cases}$$

For the primal form the term $\|\nabla u\|^2$ is missing, whereas for the dual form the missing term is $\|\nabla \cdot \mathbf{q}\|^2$. Moreover, in both cases the term $\|u\|^2$ is multiplied by σ , and we also need to account for the case $\sigma = 0$. It is thus clear that B is not coercive in the norm $\|\cdot\|_X$. However, coercivity is a sufficient condition for the problem to be well posed (i.e., to have a unique solution that can be bounded in terms of the data, according to Hadamard's definition of well posedness), but it is not a necessary condition. The necessary and sufficient condition for a variational problem to be well posed is stated next.

Let us consider an abstract problem of the form:

$$\text{Find } U \in X_U \text{ such that } B(U, V) = L(V) \quad \forall V \in X_V, \quad (1.70) \quad \{\text{eq:c1-}$$

where X_U and X_V are Banach spaces, X_V being reflexive ($X_V = X_V''$), B is a continuous bilinear form on $X_U \times X_V$ and L a continuous form on X_V . Let N_B and N_L be the norms of B and L , respectively.

Theorem 8 (Babuška-Lax-Milgram) *Under the conditions stated for problem (1.70), the necessary and sufficient conditions for this problem to be well posed are:*

$$\inf_{U \in X_U \setminus \{0\}} \sup_{V \in X_V \setminus \{0\}} \frac{|B(U, V)|}{\|U\|_{X_U} \|V\|_{X_V}} \geq K_B > 0, \quad (1.71)$$

$$\forall V \in X_V \setminus \{0\} \quad \sup_{U \in X_U \setminus \{0\}} \frac{|B(U, V)|}{\|U\|_{X_U}} > 0. \quad (1.72)$$

If these two conditions hold, the unique solution of problem (1.70) can be bounded as

$$\|U\|_{X_U} \leq \frac{N_L}{K_B}. \quad (1.73)$$

Proof. Let us start noting three different equivalent ways of writing condition (1.71):

$$\begin{aligned} & \inf_{U \in X_U \setminus \{0\}} \sup_{V \in X_V \setminus \{0\}} \frac{|B(U, V)|}{\|U\|_{X_U} \|V\|_{X_V}} \geq K_B > 0 \\ \iff & \forall U \in X_U \quad \sup_{V \in X_V \setminus \{0\}} \frac{|B(U, V)|}{\|V\|_{X_V}} \geq K_B \|U\|_{X_U} \\ \iff & \forall U \in X_U \quad \exists V \in X_V \text{ such that } |B(U, V)| \geq K_B \|U\|_{X_U} \|V\|_{X_V}. \end{aligned} \quad (1.74)$$

These three equivalences are easily checked. This condition, together with the so called non-degeneracy condition (1.72), allow one to prove existence of solution. The proof is based on the closed range theorem of functional analysis and details are outside the scope of these notes (if X_U and X_V are Hilbert spaces, existence can be alternatively proved using Riesz's representation theorem). From (1.74) one can easily show that solutions satisfy the bound (1.73), as for any solution $U \in X_U$ there exists $V \in X_V$ such that

$$K_B \|U\|_{X_U} \|V\|_{X_V} \leq |B(U, V)| = |L(V)| \leq N_L \|V\|_{X_V}.$$

This same bound serves to prove uniqueness, as if there are two solutions $U_1, U_2 \in X_U$, their difference $U_1 - U_2$ solves the homogenous counterpart of (1.70), and thus $\|U_1 - U_2\|_{X_U} \leq 0$ according to (1.73), from where $U_1 = U_2$. $\square$

The key point is condition (1.71), since condition (1.72) essentially states that X_V is not 'orthogonal' (in a certain sense) to X_U with respect to B . In particular, (1.72) is obvious if $X_U = X_V$. We will refer to (1.71) as **the inf-sup condition**. Sometimes it is referred to as **weak coercivity**, whereas the coercivity introduced earlier is called **strong coercivity**. Often times, the term *coercivity* is also replaced by **ellipticity** (weak or strong).

The name *Babuška-Lax-Milgram* chosen for the last result is probably the most common one, as it can be understood as a generalisation of Lax-Milgram's lemma to non-coercive bilinear forms in Banach spaces, and the formulation we have followed is due to Babuška. However, it is a direct consequence of previous results established by Banach and Nečas, and this is sometimes acknowledged in the name of the theorem.

Remark 1 That condition (1.71) is absolutely natural can be understood if we consider its finite dimensional version. Let $N < \infty$ be the dimension of the problem. In this case, the bilinear form B can be represented by a matrix $\mathbf{B} \in \mathbb{R}^{N \times N}$ and U and V by vectors $\mathbf{U} \in \mathbb{R}^N$ and $\mathbf{V} \in \mathbb{R}^N$, respectively, so that $B(U, V) = \mathbf{V}^T \mathbf{B} \mathbf{U}$. Condition (1.71) then reads:

$$\min_{\mathbf{U} \in \mathbb{R}^N \setminus \{0\}} \max_{\mathbf{V} \in \mathbb{R}^N \setminus \{0\}} \frac{|\mathbf{V}^T \mathbf{B} \mathbf{U}|}{\|\mathbf{U}\| \|\mathbf{V}\|} \geq K_B > 0. \quad (1.75)$$

It can be shown that this implies that all singular values of $\mathbf{B}$ are non-zero and the smallest is greater or equal to K_B . Thus, in the finite dimensional case condition (1.71) simply states that matrix $\mathbf{B}$ is non-singular. This is in contrast with coercivity, which requires that all eigenvalues of the matrix be positive. Clearly, coercivity is sufficient for solvability, but not necessary. The necessary and sufficient condition is (1.75).

Let us also remark why the previous theorem cannot be proved using the strategy employed for Lax-Milgram's lemma (theorem 1), even if $X_U = X_V \equiv X$ are separable Hilbert spaces. Let X_N be a subspace of X of dimension N . We could try to find a solution $U_N \in X_N$, but we do not have any guarantee that this problem will be well posed, because the *inf-sup condition on X is not necessarily inherited in X_N* . Suppose we are given $U_N \in X_N$; we know that there exists $V \in X$ such that (1.74) holds, but we cannot ensure that $V \in X_N$. If this holds, we can prove a generalisation of Céa's lemma:

Theorem 9 (Generalised Céa's lemma) Consider problem (1.70) with $X_U = X_V \equiv X$ and suppose that the assumptions of Theorem 8 hold. Suppose also that there exists a subspace $X_N \subset X$ for which for all $U_N \in X_N$ there exists $V_N \in X_N$ such that $K_B \|U_N\|_X \|V_N\|_X \leq |B(U_N, V_N)|$, and consider the problem of finding $U_N \in X_N$ such that $B(U_N, V_N) = L(V_N)$ for all $V_N \in X_N$. Then, this problem is well posed and there holds:

$$\|U - U_N\|_X \leq \left(1 + \frac{N_B}{K_B}\right) \inf_{W_N \in X_N} \|U - W_N\|_X. \quad (1.76)$$

Proof. Well posedness of the problem posed in X_N is a consequence of the inf-sup condition in this subspace. We will discuss it in detail when considering the FE approximation. To prove estimate (1.76), let us start noting that, since $X_N \subset X$, $B(U, V_N) = L(V_N)$ for all $V_N \in X_N$, and therefore $B(U - U_N, V_N) = 0$ for all $V_N \in X_N$ (Garlerkin orthogonality). For any $W_N \in X_N$, because of the inf-sup condition in X_N there exists $V_N \in X_N$ such that

$$\begin{aligned} K_B \|U_N - W_N\|_X \|V_N\|_X &\leq |B(U_N - W_N, V_N)| \\ &= |B(U_N - W_N, V_N) + B(U - U_N, V_N)| \\ &= |B(U - W_N, V_N)| \\ &\leq N_B \|U - W_N\|_X \|V_N\|_X, \end{aligned}$$

from where

$$\|U_N - W_N\|_X \leq \frac{N_B}{K_B} \|U - W_N\|_X.$$

From this and the triangle inequality we have:

$$\|U - U_N\|_X \leq \|U - W_N\|_X + \|U_N - W_N\|_X \leq \left(1 + \frac{N_B}{K_B}\right) \|U - W_N\|_X,$$

from where the result follows. $\square$

Theorems 8 and 9 are applicable to the general problem (1.70). However, Darcy's problem has a particular structure shared by other problems that can be exploited, which is given by Eqs. (1.58)-(1.59). We have already seen that this structure is the same for the primal and the dual forms of Darcy's problem.

In order to use the notation usually found in the literature, let us consider the following problem, with arbitrary forms and function spaces, not corresponding to those introduced earlier: find $[u, p] \in V \times Q$ such that

$$\{\text{eq:c1-sad1}\} \quad a(u, v) + b(p, v) = l_u(v), \quad (1.77)$$

$$\{\text{eq:c1-sad2}\} \quad -b(q, u) + c(p, q) = l_p(q), \quad (1.78)$$

for all $[v, q] \in V \times Q$, where V and Q are Hilbert spaces, $a : V \times V \rightarrow \mathbb{R}$, $b : Q \times V \rightarrow \mathbb{R}$ and $c : Q \times Q \rightarrow \mathbb{R}$ are bilinear forms and $l_u : V \rightarrow \mathbb{R}$ and $l_p : Q \rightarrow \mathbb{R}$ are linear forms. Problem (1.77)-(1.78) can be cast as a single variational equation of the form (1.70) by defining $X = V \times Q$ and

$$\{\text{eq:c1-B1bb}\} \quad B([u, p], [v, q]) = a(u, v) + b(p, v) - b(q, u) + c(p, q), \quad (1.79)$$

$$\{\text{eq:c1-L1bb}\} \quad L([v, q]) = l_u(v) + l_p(q). \quad (1.80)$$

It can be shown that problem (1.77)-(1.78) has the structure of a **saddle point problem** when a is positive definite and c is positive semi-definite. Indeed, we have that

Proposition 3 Suppose that in problem (1.77)-(1.78) the bilinear form a is such that $a(v, v) > 0$ for all $0 \neq v \in V$, and $c(q, q) \geq 0$ for all $q \in Q$. Then, the solution $[u, p] \in V \times Q$ of this problem satisfies:

$$\{\text{eq:c1-minmax}\} \quad [u, p] = \arg \inf_{v \in V} \sup_{q \in Q} L(v, q), \quad (1.81)$$

where

$$\{\text{eq:c1-lag}\} \quad L(v, q) := \frac{1}{2}a(v, v) + b(q, v) - \frac{1}{2}c(q, q) - l_u(v) + l_p(q). \quad (1.82)$$

Proof. Consider the function

$$f(\epsilon_1, \epsilon_2) := L(u + \epsilon_1 v, p + \epsilon_2 q),$$

where $[u, p]$ is the solution of problem (1.77)-(1.78) and $[v, q] \in V \times Q$ is arbitrary. It is immediately checked that

$$\begin{aligned} \left. \frac{\partial f}{\partial \epsilon_1} \right|_{\epsilon_1=\epsilon_2=0} &= a(u, v) + b(p, v) - l_u(v), \\ \left. \frac{\partial f}{\partial \epsilon_2} \right|_{\epsilon_1=\epsilon_2=0} &= b(q, u) - c(p, q) + l_p(q). \end{aligned}$$

From this, it follows that $[u, p]$ is a stationary point of $L(v, q)$ if, and only if, $[u, p]$ is the solution of problem (1.77)-(1.78). To show (1.82), it only remains to compute the eigenvalues of the Hessian of $f(\epsilon_1, \epsilon_2)$, which can be obtained from:

$$\det \begin{bmatrix} \frac{\partial^2 f}{\partial \epsilon_1^2} - \lambda & \frac{\partial^2 f}{\partial \epsilon_1 \partial \epsilon_2} \\ \frac{\partial^2 f}{\partial \epsilon_1 \partial \epsilon_2} & \frac{\partial^2 f}{\partial \epsilon_2^2} - \lambda \end{bmatrix} = \det \begin{bmatrix} a(v, v) - \lambda & b(q, v) \\ b(q, v) & -c(q, q) - \lambda \end{bmatrix} = 0,$$

$$\begin{aligned}\lambda^2 + (c(q, q) - a(v, v))\lambda - b(q, v)^2 - a(v, v)c(q, q) &= 0, \\ 2\lambda_{1,2} &= a(v, v) - c(q, q) \pm [(a(v, v) + c(q, q))^2 + 4b(q, v)^2]^{1/2}.\end{aligned}$$

The first eigenvalue verifies

$$\begin{aligned}2\lambda_1 &= a(v, v) - c(q, q) + [(a(v, v) + c(q, q))^2 + 4b(q, v)^2]^{1/2} \\ &\geq a(v, v) - c(q, q) + a(v, v) + c(q, q) = 2a(v, v) > 0,\end{aligned}$$

whereas for the second we have

$$\begin{aligned}2\lambda_2 &= a(v, v) - c(q, q) - [(a(v, v) + c(q, q))^2 + 4b(q, v)^2]^{1/2} \\ &< -2c(q, q) \leq 0.\end{aligned}$$

Thus, one eigenvalue is positive and the other is negative, from where the saddle point structure given by Eq. (1.81) follows. $\square$

Furthermore, in the particular case in which $c = 0$ and $l_p = 0$, the variable p can be understood as the **Lagrange multiplier** to enforce the condition $A_b^T u = 0$, where $A_b^T : V \rightarrow Q'$ is defined by $A_b^T v = b(\cdot, v)$, in the minimisation of the functional $\frac{1}{2}a(v, v) - l_u(v)$. This is why often the term *saddle point problem* is restricted to the case in which $c = 0$, and when c is introduced one refers to a **perturbed saddle point problem**. We shall come back to this discussion when studying the Stokes problem.

For problem (1.77)-(1.78) the expression of the inf-sup condition can be particularised. In fact, we shall only consider sufficient conditions to guarantee that the bilinear form in Eq. (1.79) satisfies an inf-sup condition:

Theorem 10 (Ladyzhenskaya-Babuška-Brezzi) *Suppose that*

H1 The bilinear forms a and b are continuous, with continuity constants N_a and N_b .

H2 The bilinear form a is symmetric and non-negative in V , and coercive in the subspace

$$K := \{v \in V \mid b(q, v) = 0 \ \forall q \in Q\},$$

i.e., there exists $K_a > 0$ such that $a(u, u) \geq K_a \|u\|_V^2$ for all $u \in K$.

H3 The bilinear form b satisfies the following inf-sup condition: there exists $K_b > 0$ such that

$$\forall p \in Q \ \exists v \in V \mid K_b \|p\|_Q \|v\|_V \leq b(p, v). \quad (1.83) \quad \{\text{eq:c1-lbb}\}$$

H4 The bilinear form c is continuous and coercive in a norm $\|\cdot\|_{\hat{Q}}$, possibly weaker than $\|\cdot\|_Q$, i.e., there exists $N_c \geq 0$ and $K_c = \gamma N_c$, with $\gamma > 0$, such that $c(p, q) \leq N_c \|p\|_{\hat{Q}} \|q\|_{\hat{Q}}$ and $c(p, p) \geq K_c \|p\|_{\hat{Q}}^2$ for all $p, q \in Q$.

Then, the bilinear form B in (1.79) is continuous and satisfies the inf-sup condition

$$\forall [u, p] \in V \times Q \ \exists [v, q] \in V \times Q \mid K_B \|[u, p]\|_{V \times Q} \|[v, q]\|_{V \times Q} \leq B([u, p], [v, q]), \quad (1.84) \quad \{\text{eq:c1-lbb}\}$$

for a constant $K_B > 0$, where

$$\|[v, q]\|_{V \times Q} := \sqrt{K_a} \|v\|_V + \sqrt{M_{ab}} \|q\|_Q. \quad (1.85) \quad \{\text{eq:c1-pro}\}$$

where $M_{ab} > 0$ depends on K_a , K_b , and N_a .

Proof. The continuity of B is trivial, and its norm can be easily written in terms of the norms N_a , N_b and N_c and, because of definition (1.85), also of K_a .

Let us prove the inf-sup condition for B . First, since V is a Hilbert space we can consider the decomposition $V = K \oplus K^\perp$, where $K^\perp$ is the orthogonal to K with respect to any inner product in V that restricted to K coincides with $a(\cdot, \cdot)$. This allows us to split any $v \in V$ as $v = v_0 + v_1$, with $v_0 \in K$ and $v_1 \in K^\perp$, and $a(v, v) = a(v_0, v_0) + a(v_1, v_1)$. Let $u = u_0 + u_1 \in V$ and $p \in Q$ be given. Let us obtain $[v, q] \in V \times Q$ for which (1.84) holds.

In a first step, observe that:

$$B([u, p], [u_0, 0]) = a(u_0, u_0) \geq K_a \|u_0\|_V^2. \quad (1.86)$$

For the second step we need to use a property that follows from the closed range theorem of functional analysis, which states that if (1.83) holds, then

$$\forall v \in K^\perp \exists q \in Q \mid K_b \|q\|_Q \|v\|_V \leq b(q, v),$$

i.e., the arguments in the inf-sup condition can be switched when $v \in K^\perp$ and the inf-sup constant is the same (in the finite dimensional case, this amounts to saying that the rank of a matrix is equal to the rank of its transpose and the singular values are the same). This in particular applies to $u_1 \in K^\perp$. Let q_1 be the element in Q for which this inequality holds; the norm of q_1 is arbitrary, and can be taken as $\|q_1\|_Q = \|u_1\|_V K_a / K_b$. We have that

$$\begin{aligned} B([u, p], [0, q_1]) &= -b(q_1, u_1) + c(p, q_1) \\ &\geq K_b \|q_1\|_V \|u_1\|_V - N_c \|p\|_{\hat{Q}} \|q_1\|_Q && \text{Since } \|q_1\|_{\hat{Q}} \leq \|q_1\|_Q \\ &\geq K_a \|u_1\|_V^2 - \left(\frac{1}{2\alpha} \|p\|_{\hat{Q}}^2 + \frac{\alpha}{2} \|u_1\|_V^2 \right) \frac{N_c K_a}{K_b} && \text{Using Young's inequality} \\ &\geq \frac{K_a}{2} \|u_1\|_V^2 - \frac{N_c^2 K_a}{2K_b^2} \|p\|_{\hat{Q}}^2 && \text{Taking } \alpha = \frac{K_b}{N_c} \text{ if } N_c \neq 0. \end{aligned} \quad (1.87)$$

If $N_c = 0$ the result is obviously true, also.

The third step is to use the inf-sup condition (1.83). If, given $p \in Q$, v_p is the element in V for which it holds and we take its norm as $\|v_p\|_V = \frac{K_a K_b}{N_a^2} \|p\|_Q$, we have that

$$\begin{aligned} B([u, p], [v_p, 0]) &= a(u, v_p) + b(p, v_p) \\ &\geq K_b \|p\|_Q \|v_p\|_V - N_a \|v_p\|_V \|u\|_V \\ &\geq K_b \|p\|_Q \|v_p\|_V - \frac{N_a}{2\alpha} \|u\|_V^2 - \frac{N_a \alpha}{2} \|v_p\|_V^2 && \text{Using Young's inequality} \\ &\geq \frac{K_a K_b^2}{2N_a^2} \|p\|_Q^2 - \frac{K_a}{2} \|u\|_V^2 && \text{Taking } \alpha = \frac{N_a}{K_a} \end{aligned} \quad (1.88)$$

Finally, we have that

$$B([u, p], [u, p]) = a(u, u) + c(p, p) \geq K_c \|p\|_Q^2. \quad (1.89)$$

Let $[v, q] = [u_0 + v_p + \beta u, 2q_1 + \beta p]$. From Eq. (1.86)-(1.89) we obtain:

$$B([u, p], [v, q]) \gtrsim K_a \|u\|_V^2 + \frac{K_a K_b^2}{N_a^2} \|p\|_Q^2 + \left(\beta \gamma N_c - \frac{N_c^2 K_a}{K_b^2} \right) \|p\|_Q^2. \quad (1.90)$$

Taking β large enough it follows that $B([u, p], [v, q]) \gtrsim \|[u, p]\|_{V \times Q}^2$. The proof concludes checking that $\|[v, q]\|_{V \times Q} \lesssim \|[u, p]\|_{V \times Q}$. $\square$

Object	Abstract setting	Darcy primal	Darcy dual
Unknown	$U = [u, p]$	$U = [\mathbf{q}, u]$	$U = [\mathbf{q}, u]$
Test function	$V = [v, q]$	$V = [\mathbf{r}, v]$	$V = [\mathbf{r}, v]$
Spaces	$u \in V$ $q \in Q$	$\mathbf{q} \in R = L^2(\Omega)^d$ $v \in V = \{v \in H^1(\Omega) \mid v = 0 \text{ on } \Gamma_u\}$	$\mathbf{q} \in R = \{\mathbf{r} \in H(\text{div}, \Omega) \mid \mathbf{n} \cdot \mathbf{r} = 0 \text{ on } \Gamma_g \text{ and } \mathbf{n} \cdot \mathbf{r} \in L^2(\Gamma_u) \text{ if } \beta_u \neq 0\}$ $v \in V = L^2(\Omega)$
Bilinear forms	$a(u, v)$ $b(q, v)$ $c(p, q)$	$a(\mathbf{q}, \mathbf{r}) = \kappa^{-1}(\mathbf{q}, \mathbf{r})$ $b(v, \mathbf{r}) = (\nabla v, \mathbf{r})$ $c(u, v) = \sigma(v, u) + \beta_g(v, u)_{L^2(\Gamma_g)}$	$a(\mathbf{q}, \mathbf{r}) = \kappa^{-1}(\mathbf{q}, \mathbf{r}) + \beta_u(\mathbf{n} \cdot \mathbf{q}, \mathbf{n} \cdot \mathbf{r})_{\Gamma_u}$ $b(v, \mathbf{r}) = -(v, \nabla \cdot \mathbf{r})$ $c(u, v) = \sigma(v, u)$
Linear forms	$l_u(v)$ $l_p(q)$	$l_q(\mathbf{r}) = 0$ $l_u(v) := \langle v, f \rangle + \langle v, g \rangle_{\Gamma_g}$	$l_q(\mathbf{r}) := -\langle \bar{u}, \mathbf{n} \cdot \mathbf{r} \rangle_{\Gamma_u}$ $l_u(v) := (v, f)$
Norms	$\ v\ _V^2$ $\ q\ _Q^2$ $\ q\ _{\hat{Q}}^2$	$\kappa^{-1}\ \mathbf{r}\ ^2$ $\kappa \ell_\Omega^{-2}\ v\ ^2 + \kappa\ \nabla v\ ^2$ $\sigma\ v\ ^2 + \beta_g\ v\ _{L^2(\Gamma_g)}^2$	$\kappa^{-1}\ \mathbf{r}\ ^2 + \kappa^{-1}\ell_\Omega^2\ \nabla \cdot \mathbf{r}\ ^2$ $\kappa \ell_\Omega^{-2}\ v\ ^2$ $\sigma\ v\ ^2$
Constants	K_a, N_a K_b	$K_a = N_a = \kappa^{-1}$ ℓ_Ω^{-1}	$K_a = N_a = \kappa^{-1}$ ℓ_Ω^{-1}

Table 1.1: Identification of Darcy's problems (primal and dual) with the abstract mixed problem

{tab:c1-t1

The proof presented is a non-standard proof of Ladyzhenskaya-Babuška-Brezzi's Theorem. It allows us to check how the constants depend on the norm N_a and the constants K_a and K_b . It can be generalised in several ways, for example requiring only an inf-sup condition of $a(\cdot, \cdot)$ in K instead of coercivity. Symmetry of $a(\cdot, \cdot)$ can also be relaxed, as we have only used it to construct $K^\perp$, but it could be built by other means. Referring to the bilinear form $c(\cdot, \cdot)$, a typical example is $c(p, q) = \lambda(p, q)_Q$, with $\lambda \geq 0$, case in which $K_c = N_c = \lambda$ and $\|\cdot\|_{\hat{Q}} = \|\cdot\|_Q$. In fact, if K_c is 'large', the problem is well posed without the need for H3, as stability for p can be obtained from $c(\cdot, \cdot)$.

It is obvious that this result applies to the finite dimensional case as well, i.e., to the discrete problem. Thus, if assumptions H1 to H4 can be checked, Theorem 9 provides a best approximation property of the discrete solution to the continuous one. If these assumptions hold for the continuous problem, all but H3 carry over to the discrete problem for conforming FE approximations (for H2, note that if $v_h \in V_h \cap K$, then $v_h \in K_h = \{v_h \in V_h \mid b(q_h, v_h) = 0 \forall q_h \in Q_h \subset Q\}$). Therefore, the key in the FE approximation of mixed saddle point problems is to check H3.

Remark 2 *The statement of the theorem presented is essentially due to Brezzi, but as it has been proved it is a particular case of Theorem 8, whose statement is due Babuška. Ladyzhenskaya proved earlier it holds for the Stokes problem (all this at the continuous level), and hence the name of the theorem. The key condition H3 is often called Ladyzhenskaya-Babuška-Brezzi's condition, or simply **LBB condition**. Its discrete version is critical in the FE approximation of the problem.*

1.4.5 Finite element approximation of Darcy's problem

First of all, let us start collecting the definitions introduced up to now, both for Darcy's problem (in primal and dual forms) and for the abstract problem for which Theorems 8, 9 and 10 have been stated. This is done in Table 1.1. This table has to be read with care, as the same symbol can have different meanings depending on the context. The expression of K_a , N_a and K_b is given up to dimensionless constants; K_a and N_a follow from the expression of the bilinear form a , whereas K_b can be justified from a scaling argument.

Let us consider a conforming FE approximation $R_h \times V_h$ of space $R \times V$, with $R_h \subset R$ and $V_h \subset V$. As it has been said previously, it is not the objective of these notes to construct particular FE spaces. However, it is important to know examples of spaces R_h and V_h that satisfy the inf-sup condition given by Eq. (1.83), that we shall call **inf-sup stable pairs**; for these, Theorem 10 allows us to use the general approximation Theorem 9. The critical point is to check whether assumption H3 (expressed by (1.83)) is satisfied or not. Again, we will not explain the techniques employed to prove that this condition is satisfied for a particular FE pair $R_h \times V_h$.

Primal formulation

In the case of the primal formulation, it is easy to construct inf-sup stable pairs. In this case, $R_h \subset R = L^2(\Omega)^d$ and $V_h \subset V \subset H^1(\Omega)$. The inf-sup condition (1.83) takes the form

$$\forall v_h \in V_h \exists \mathbf{r}_h \in R_h \mid K_b \|v_h\|_{H^1(\Omega)} \|\mathbf{r}_h\|_{L^2(\Omega)} \leq (\nabla v_h, \mathbf{r}_h),$$

for a constant $K_b > 0$. The easiest way to construct $R_h \times V_h$ is to start building V_h as a conforming space of V (using for example Lagrangian elements) and then take R_h as the space of gradients of functions in V_h . Thus, if V_h is made of continuous polynomials of degree k on simplicial elements, R_h would be made of piecewise discontinuous polynomials of degree $k-1$ in each component. The inf-sup condition in this case is trivial to check, as given $v_h \in V_h$ we may take $\mathbf{r}_h = \nabla v_h \in R_h$ and then

$$(\nabla v_h, \mathbf{r}_h) = \|\nabla v_h\|^2 \geq \ell_\Omega^{-1} \|\nabla v_h\| \|v_h\|_{H^1(\Omega)} = \ell_\Omega^{-1} \|\mathbf{r}_h\| \|v_h\|_{H^1(\Omega)},$$

where the Poincaré-Friedrichs inequality has been used. Thus, the discrete version of the inf-sup condition (1.83) holds with $K_b = \ell_\Omega^{-1}$. This case, however, is of no practical interest, as the FE solution to Darcy's problem would be exactly the same as the solution to the irreducible problem. The approximation of the primal formulation may be of interest in cases in which κ is variable (not considered here) and using *equal* interpolation for u_h and $\mathbf{q}_h$. However, the resulting FE spaces V_h and R_h *do not satisfy the inf-sup condition in this case*. We will see that they can be employed when stabilised FE formulations are used.

This discussion serves to justify that, in general, the interpolation for the different variables in a mixed problem will be different. These are precisely called **mixed FE interpolations**. When the problem has two unknowns, one has to consider two different sets of interpolation functions and two sets of degrees of freedom, one for each variable. The subdomains of the FE partition of Ω will be the only common ingredient for the two unknowns. Let us explain how to obtain the discrete system in a general case, including the primal and the dual formulations.

The two FE spaces R_h and V_h can be constructed in a way similar to the FE space constructed for the irreducible problem. In particular, one has to start with the broken FE spaces

$$\begin{aligned} R_{h,k_q,q} &= \left\{ \mathbf{r}_h : \Omega \rightarrow \mathbb{R}^d \mid \mathbf{r}_h|_K = \hat{\mathbf{r}} \circ \boldsymbol{\varphi}_K^{-1}, \hat{\mathbf{r}} \in R_{k_q,q}(\hat{K}), K \in \mathcal{T}_h \right\}, \\ R_{h,k_u,u} &= \left\{ v_h : \Omega \rightarrow \mathbb{R} \mid v_h|_K = \hat{v} \circ \boldsymbol{\varphi}_K^{-1}, \hat{v} \in R_{k_u,u}(\hat{K}), K \in \mathcal{T}_h \right\}. \end{aligned}$$

Note that we have considered the same diffeomorphism $\boldsymbol{\varphi}_K$ for the two spaces, as the subdomain $K \in \mathcal{T}_h$ is the same for both. The degree of the polynomials in $R_{h,k_q,q}$ and $R_{h,k_u,u}$ can be different, and in fact one could be $P_k(\hat{K})$ and the other $Q_k(\hat{K})$, for $k = k_u$ or $k = k_q$.

Now one needs to construct the local spaces $R_{k_q,q}(\hat{K})$ and $R_{k_u,u}(\hat{K})$, and this involves two sets of shape (basis) functions and two sets of degrees of freedom (one for $\mathbf{q}_h$ and another one for u_h) with $n_{\text{lo},q} = \dim(R_{h,k_q,q})$ and $n_{\text{lo},u} = \dim(R_{h,k_u,u})$ components each, respectively. Thus:

$$R_{k_q,q}(\hat{K}) = \text{span}\{\hat{\mathbf{N}}_q^1(\boldsymbol{\xi}), \dots, \hat{\mathbf{N}}_q^{n_{\text{lo},q}}(\boldsymbol{\xi})\},$$

$$R_{k_u,u}(\hat{K}) = \text{span}\{\hat{N}_u^1(\boldsymbol{\xi}), \dots, \hat{N}_u^{n_{\text{Ino},u}}(\boldsymbol{\xi})\},$$

and

$$\hat{\mathbf{r}}(\boldsymbol{\xi}) = \sum_{A=1}^{n_{\text{Ino},q}} \hat{N}_q^A(\boldsymbol{\xi}) \alpha_q^A, \quad \hat{\mathbf{v}}(\boldsymbol{\xi}) = \sum_{A=1}^{n_{\text{Ino},u}} \hat{N}_u^A(\boldsymbol{\xi}) \alpha_u^A,$$

for two sets of degrees of freedom $\{\alpha_q^A\}_{A=1}^{n_{\text{Ino},q}}$ and $\{\alpha_u^A\}_{A=1}^{n_{\text{Ino},u}}$. Note that the basis functions for $R_{k_q,q}(\hat{K})$ are vector fields. Once the basis functions are constructed in the reference domain $\hat{K}$ we can obtain them in each domain K of the FE partition as

$$\begin{aligned} N_{K,q}^A(\mathbf{x}) &= \hat{N}_q^A(\boldsymbol{\varphi}_K^{-1}(\mathbf{x})), \quad A = 1, \dots, n_{\text{Ino},q}, \quad \mathbf{x} \in K, \\ N_{K,u}^A(\mathbf{x}) &= \hat{N}_u^A(\boldsymbol{\varphi}_K^{-1}(\mathbf{x})), \quad A = 1, \dots, n_{\text{Ino},u}, \quad \mathbf{x} \in K. \end{aligned}$$

These local shape functions allow us to write

$$\mathbf{r}_h(\mathbf{x})|_K = \sum_{A=1}^{n_{\text{Ino},q}} N_{K,q}^A(\mathbf{x}) \alpha_{K,q}^A, \quad v_h(\mathbf{x})|_K = \sum_{A=1}^{n_{\text{Ino},u}} N_{K,u}^A(\mathbf{x}) \alpha_{K,u}^A.$$

Once the unknowns are interpolated locally with piecewise polynomials (up to the composition with $\boldsymbol{\varphi}_K^{-1}$) we can construct a global interpolation for each of them, of the form

$$\mathbf{r}_h(\mathbf{x}) = \sum_{a=1}^{n_{\text{gno},q}} N_q^a(\mathbf{x}) \alpha_q^a, \quad v_h(\mathbf{x}) = \sum_{a=1}^{n_{\text{gno},u}} N_u^a(\mathbf{x}) \alpha_u^a,$$

where $n_{\text{gno},q}$ is the total number of degrees of freedom for $\mathbf{q}_h$ and $n_{\text{gno},u}$ for u_h .

At this point the discrete variational equations can be obtained exactly as for the irreducible case. Let us write the discrete version of problem (1.60) as: find $[\mathbf{q}_h, u_h] \in R_h \times V_h$ such that

$$a(\mathbf{q}_h, \mathbf{r}_h) + b(u_h, \mathbf{r}_h) - b(v_h, \mathbf{q}_h) + c(u_h, v_h) = l_q(\mathbf{r}_h) + l_u(v_h) \quad \forall [\mathbf{r}_h, v_h] \in R_h \times V_h. \quad (1.91) \quad \{\text{eq:c1-mm1}\}$$

Suppose that the unknowns are written as

$$\mathbf{q}_h(\mathbf{x}) = \sum_{a=1}^{n_{\text{gno},q}} N_q^a(\mathbf{x}) \beta_q^a, \quad u_h(\mathbf{x}) = \sum_{A=1}^{n_{\text{gno},u}} N_u^A(\mathbf{x}) \beta_u^A,$$

and let us organise the degrees of freedom of $\mathbf{q}_h$, u_h and the generic test functions $\mathbf{r}_h$, v_h in the arrays

$$\begin{aligned} \mathbf{q}^T &= (\beta_q^1, \dots, \beta_q^{n_{\text{gno},q}}), \quad \mathbf{r}^T = (\alpha_q^1, \dots, \alpha_q^{n_{\text{gno},q}}), \\ \mathbf{u}^T &= (\beta_u^1, \dots, \beta_u^{n_{\text{gno},u}}), \quad \mathbf{v}^T = (\alpha_u^1, \dots, \alpha_u^{n_{\text{gno},u}}), \end{aligned}$$

where $n_{\text{gno},q}$ is the number of degrees of freedom of $\mathbf{q}_h$ excluding prescriptions coming from boundary conditions and $n_{\text{gno},u}$ is likewise for u_h . Then, problem (1.91) can be written as

$$\mathbf{r}^T \mathbf{A}_a \mathbf{q} + \mathbf{r}^T \mathbf{A}_b \mathbf{u} - \mathbf{v}^T \mathbf{A}_b^T \mathbf{q} + \mathbf{v}^T \mathbf{A}_c \mathbf{u} = \mathbf{r}^T \mathbf{f}_q + \mathbf{v}^T \mathbf{f}_u,$$

for all arrays $\mathbf{r} \in \mathbb{R}^{n_{\text{gno},q}}$ and $\mathbf{v} \in \mathbb{R}^{n_{\text{gno},u}}$, which yields the algebraic system

$$\begin{bmatrix} \mathbf{A}_a & \mathbf{A}_b \\ -\mathbf{A}_b^T & \mathbf{A}_c \end{bmatrix} \begin{bmatrix} \mathbf{q} \\ \mathbf{u} \end{bmatrix} = \begin{bmatrix} \mathbf{f}_q \\ \mathbf{f}_u \end{bmatrix}, \quad (1.92) \quad \{\text{eq:c1-mam}\}$$

where

$$\begin{aligned} \mathbf{A}_a &= [\mathbf{A}_a^{ab}] \in \mathbb{R}^{n_{\text{gnf},q} \times n_{\text{gnf},q}}, & \mathbf{A}_a^{ab} &= a(\mathbf{N}_q^b, \mathbf{N}_q^a), \\ \mathbf{A}_b &= [\mathbf{A}_b^{ab}] \in \mathbb{R}^{n_{\text{gnf},q} \times n_{\text{gnf},u}}, & \mathbf{A}_b^{ab} &= b(\mathbf{N}_u^b, \mathbf{N}_q^a), \\ \mathbf{A}_c &= [\mathbf{A}_c^{ab}] \in \mathbb{R}^{n_{\text{gnf},u} \times n_{\text{gnf},u}}, & \mathbf{A}_c^{ab} &= c(\mathbf{N}_u^b, \mathbf{N}_u^a), \\ \mathbf{f}_q &= [\mathbf{f}_q^a] \in \mathbb{R}^{n_{\text{gnf},q}}, & \mathbf{f}_q^a &= l_q(\mathbf{N}_q^a), \\ \mathbf{f}_u &= [\mathbf{f}_u^a] \in \mathbb{R}^{n_{\text{gnf},u}}, & \mathbf{f}_u^a &= l_u(\mathbf{N}_u^a). \end{aligned}$$

The simplest example of inf-sup stable interpolation in the case of the primal formulation and for $d = 2$ would be obtained constructing V_h from piecewise continuous polynomials of degree one and R_h from piecewise constant vector fields on each element domain. In this case we would have $k_u = 1$, $k_q = 0$, $n_{\text{Ino},u} = 3$, $n_{\text{Ino},q} = 2$ (two components of $\mathbf{q}_h$), $n_{\text{gnf},u}$ equal to the number of points (up to boundary conditions) and $n_{\text{gnf},q}$ equal to the twice the number of elements of the FE mesh. We could consider a Lagrangian interpolation also for R_h , with only one node per element (located anywhere) and with shape function for each vector component given by the characteristic function of the element domain.

Finally, if R_h is made of polynomials of degree k_q and V_h of polynomials of degree k_u , the pair $R_h \times V_h$ satisfying the discrete version of the inf-sup condition (1.83), the previous results yield the following error estimate under h -refinement for smooth enough solutions:

$$\begin{aligned} &\kappa^{-1/2} \|\mathbf{q} - \mathbf{q}_h\| + \kappa^{1/2} \ell_\Omega^{-1} \|u - u_h\| + \kappa^{1/2} \|\nabla u - \nabla u_h\| \\ &\lesssim \inf_{\mathbf{r}_h \in R_h} \kappa^{-1/2} \|\mathbf{q} - \mathbf{r}_h\| + \inf_{v_h \in V_h} \left(\kappa^{1/2} \ell_\Omega^{-1} \|u - v_h\| + \kappa^{1/2} \|\nabla u - \nabla v_h\| \right) \\ &\lesssim \kappa^{-1/2} h^{k_q+1} |\mathbf{q}|_{H^{k_q+1}(\Omega)} + \kappa^{1/2} \ell_\Omega^{-1} h^{k_u+1} |u|_{H^{k_u+1}(\Omega)} + \kappa^{1/2} h^{k_u} |u|_{H^{k_u+1}(\Omega)}. \end{aligned}$$

It is observed that the error term in $\mathbf{q}$ and in u are of the same order if $k_q + 1 = k_u$. In this case, this estimate is of order $\mathcal{O}(h^{k_u})$. This is optimal for $\|\nabla u - \nabla u_h\|$, but not for $\|u - u_h\|$. In fact, for $\|u - u_h\|$ one can obtain an improved estimate of order $\mathcal{O}(h^{k_u+1})$ using duality arguments.

Dual formulation

In the case of the dual formulation we have that $R_h \subset H(\text{div}, \Omega)$ and $V_h = L^2(\Omega)$. In this case it is not as easy as for the primal formulation to construct inf-sup stable pairs $R_h \times V_h$. The inf-sup condition can be shown to hold at the continuous level, i.e., it can be proved that

$$\{1\text{-inf-sup-dual-c}\} \quad \forall v \in V \exists \mathbf{r} \in R \mid K_b \|v\| \|\mathbf{r}\|_{H(\text{div}, \Omega)} \leq (v, \nabla \cdot \mathbf{r}), \quad (1.93)$$

for a constant $K_b > 0$ that can be taken as $K_b = \ell_\Omega^{-1}$ (up to dimensionless constants). However, this does not imply that

$$\{-\text{inf-sup-dual-d1}\} \quad \forall v_h \in V_h \exists \mathbf{r}_h \in R_h \mid K_b \|v_h\| \|\mathbf{r}_h\|_{H(\text{div}, \Omega)} \leq (v_h, \nabla \cdot \mathbf{r}_h), \quad (1.94)$$

as given $v_h \in V_h$ the element $\mathbf{r} \in R$ that satisfies condition (1.93) does not necessarily belong to $R_h \subset R$, i.e., the continuous inf-sup condition is not inherited at the discrete level, as already explained.

Condition (1.94) holds if, and only if,

$$\{-\text{inf-sup-dual-d2}\} \quad \forall \mathbf{r}_h \in R_h \exists v_h \in V_h \mid K_b \|v_h\| \|\mathbf{r}_h\|_{H(\text{div}, \Omega)} \leq (v_h, \nabla \cdot \mathbf{r}_h). \quad (1.95)$$

As mentioned earlier, this follows from the fact that the rank of a matrix is equal to the rank of its transpose and the singular values are the same.

The inf-sup condition (1.94) is not easy to satisfy. We shall neither explain general strategies to check it (are there are seven) nor give a comprehensive list of pairs that verify it; both topics can be found in many good textbooks about the FE method. We shall only construct an example of a pair $R_h \times V_h$ that verifies this. Before that, let us remark that from the conceptual point of view, verifying this condition amounts to prove the existence of an interpolation mapping P_{R_h} such that the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\nabla \cdot} & V \\ \downarrow P_{R_h} & & \downarrow P_{V_h} \\ R_h & \xrightarrow{\nabla \cdot} & V_h \end{array} \quad (1.96) \quad \{\text{eq:c1-cd1}\}$$

where P_{V_h} can be taken as the standard $L^2(\Omega)$ projection. Let us for example prove (1.94) assuming commutativity of this diagram. From the continuous inf-sup condition (1.93) it follows that

$$\forall v_h \in V_h \exists \mathbf{r} \in R \mid K_b \|v_h\| \|\mathbf{r}\|_{H(\text{div}, \Omega)} \leq (v_h, \nabla \cdot \mathbf{r}) = (v_h, P_{V_h}(\nabla \cdot \mathbf{r})) = (v_h, \nabla \cdot P_{R_h}(\mathbf{r})),$$

from where $\mathbf{r}_h = P_{R_h}(\mathbf{r}) \in R_h$ can be taken as the element for which condition (1.94) holds.

The example of pair $R_h \times V_h$ we describe now is the well known **Raviart-Thomas element** of lowest degree on triangles ($d = 2$). In this case, space V_h is made of piecewise constant functions, i.e., $k_u = 0$ and $n_{\text{Ino},u} = 1$, and thus we may write in a compact way $V_h = P_0$. The point is how to construct R_h . The first crucial fact to state is that if we wish to have $R_h \subset R \subset H(\text{div}, \Omega)$, *vector fields in R_h must have a continuous normal component across inter-element boundaries* (recall that vector fields in $H(\text{div}, \Omega)$ have a well defined normal trace on curves in 2D or surfaces in 3D). This can be proved by formalising a simple idea. Suppose that $d = 2$, consider an interior edge of the FE partition and use as coordinates the normal and the tangent to this edge. Then the divergence of a vector field in R_h on that edge is the normal derivative of the normal component plus the tangent derivative of the tangent component. The latter is obviously in $L^2(\Omega)$, as the vector field is assumed to be polynomial on both sides of the edge. But if we wish the former to be also in $L^2(\Omega)$, it is clear that we need the normal component to be continuous, since otherwise its normal derivative would behave as a δ -distribution. The reciprocal is obviously true, if a piecewise polynomial field has continuous normal components, then it is in $H(\text{div}, \Omega)$.

Once this basic fact is set, let us construct the basis functions within an element subdomain $K \in \mathcal{T}_h$. Let $\mathbf{x}^1, \mathbf{x}^2$ and $\mathbf{x}^3$ be the coordinates of the vertices of the triangle, and let $\mathbf{x}^{AB}$ be the vector that goes from $\mathbf{x}^A$ to $\mathbf{x}^B$ with normal $\mathbf{n}^{AB}$, $A, B = 1, 2, 3$. We will consider the sense of this normal *fixed* for all the mesh, so that it is not necessarily exterior to ∂K . We then define the functions:

$$\mathbf{N}_{K,q}^A(\mathbf{x}) := \frac{\text{sgn}}{L^A}(\mathbf{x} - \mathbf{x}^A), \quad A = 1, 2, 3, \mathbf{x} \in K, \quad (1.97) \quad \{\text{eq:c1-rt}\}$$

where L^A is the length of the height from node A to the opposite edge BC , and $\text{sgn} = 1$ if $\mathbf{n}^{BC}$ is exterior to ∂K and $\text{sgn} = -1$ otherwise. These functions have two useful properties:

- $\mathbf{N}_{K,q}^A(\mathbf{x}) \cdot \mathbf{n}^{AB} = \mathbf{N}_{K,q}^A(\mathbf{x}) \cdot \mathbf{n}^{AC} = 0$ on the two edges that contain node A ($A \neq B \neq C$).
- $\mathbf{N}_{K,q}^A(\mathbf{x}) \cdot \mathbf{n}^{BC}$ is constant on edge BC opposite to node A . Indeed, we have that $\mathbf{N}_{K,q}^A(\mathbf{x}^B) \cdot \mathbf{n}^{BC} = \frac{\text{sgn}}{L^A} \mathbf{x}^{AB} \cdot \mathbf{n}^{BC} = \frac{\text{sgn}}{L^A} \mathbf{x}^{AC} \cdot \mathbf{n}^{BC} = \mathbf{N}_{K,q}^A(\mathbf{x}^C) \cdot \mathbf{n}^{BC} = 1$, and since $\mathbf{N}_{K,q}^A(\mathbf{x}) \cdot \mathbf{n}^{BC}$ is linear it will be constant along edge BC , its value being $\mathbf{N}_{K,q}^A(\mathbf{x}) \cdot \mathbf{n}^{BC} = 1$.

Thus, within each $K \in \mathcal{T}_h$ we consider the interpolation

$$\forall \mathbf{r}_h \in R_h, \mathbf{r}_h(\mathbf{x})|_K = \sum_{A=1}^3 \mathbf{N}_{K,q}^A(\mathbf{x}) \alpha_{K,q}^A,$$

i.e., $n_{\text{Ino},q} = 3$. We will have that across an edge BC of ∂K :

$$\forall \mathbf{r}_h \in R_h, \mathbf{r}_h(\mathbf{x})|_{BC} \cdot \mathbf{n}^{BC} = \mathbf{N}_{K,q}^A(\mathbf{x})|_{BC} \cdot \mathbf{n}^{BC} \alpha_{K,q}^A = \alpha_{K,q}^A, \quad A \neq B \neq C,$$

which means that the degree of freedom $\alpha_{K,q}^A$ is precisely the normal component of $\mathbf{r}_h$ on the edge opposite to node A in element K (recall that the sense of the normals has been chosen a priori). Therefore, if we take as global degrees of freedom these normal components, i.e., the same by two elements that share an edge, we have guaranteed the continuity of the normal components of the vector fields in R_h , and consequently that $R_h \subset H(\text{div}, \Omega)$.

The element just constructed is referred to as the Raviart-Thomas element of order zero, often denoted RT_0 . Higher order elements RT_k are constructed by adding complete polynomials of degree k in each component to this FE space.

It is not difficult to prove that the pair $R_h \times V_h = RT_0 \times V_0$ satisfies the inf-sup condition, i.e., that a mapping P_{R_h} can be constructed such that the diagram (1.96) is commutative. This operator is constructed as

$$\forall \mathbf{r} \in R, P_{R_h}(\mathbf{r})(\mathbf{x})|_K = \sum_{A=1}^3 \mathbf{N}_{K,q}^A(\mathbf{x}) \alpha_{K,q}^A, \quad \alpha_{K,q}^A = \frac{1}{L^{BC}} \int_{BC} \mathbf{r} \cdot \mathbf{n}^{BC},$$

where L^{BC} is the length of edge BC . Recall that $\mathbf{n}^{BC}$ does not depend on the element domain K considered. The proof that diagram (1.96) is commutative is omitted.

Remark 3 Two observations are in order:

- We have directly constructed the Raviart-Thomas element in the element domain K , not in the reference domain $\hat{K}$. It is assumed that the mapping between both is affine.
- The factor $1/L^A$ ensures that $\mathbf{N}_{K,q}^A(\mathbf{x}) \cdot \mathbf{n}^{BC} = 1$. Sometimes it is omitted, and in fact $\mathbf{x}^A$ in Eq. (1.97) is not identified with the coordinates of a node. All the construction remains valid, but then the degrees of freedom are not exactly the normal component of the unknowns on the edges.

This incursion into the construction of a particular FE space is only intended to be an example of how to construct inf-sup stable FE pairs for the dual form of Darcy's problem, but it is not the objective of these notes. Apart from the Raviart-Thomas elements, other popular spaces are the Brezzi-Douglas-Marini (BDM) ones. Both families can be constructed for triangles and quads, and both in 2D and 3D.

Once the basis functions for R_h and V_h have been defined, the matrix version of the problem can be obtained exactly as for the primal formulation. Exactly the same matrix structure given by Eq. (1.92) is obtained now, the only difference being the expression of the forms involved in the construction of the arrays in this expression.

Let us conclude with an error estimate, as done for the primal formulation. If R_h is made of polynomials of degree k_q and V_h of complete polynomials of degree k_u , the pair $R_h \times V_h$ satisfying the discrete version of the inf-sup condition (1.83), we have the following error estimate for smooth enough solutions:

$$\begin{aligned} & \kappa^{-1/2} \|\mathbf{q} - \mathbf{q}_h\| + \kappa^{-1/2} \ell_\Omega \|\nabla \cdot \mathbf{q} - \nabla \cdot \mathbf{q}_h\| + \kappa^{1/2} \ell_\Omega^{-1} \|u - u_h\| \\ & \lesssim \inf_{\mathbf{r}_h \in R_h} \left(\kappa^{-1/2} \|\mathbf{q} - \mathbf{r}_h\| + \kappa^{-1/2} \ell_\Omega \|\nabla \cdot \mathbf{q} - \nabla \cdot \mathbf{r}_h\| \right) + \inf_{v_h \in V_h} \kappa^{1/2} \ell_\Omega^{-1} \|u - v_h\| \\ & \lesssim \kappa^{-1/2} h^{k_q+1} |\mathbf{q}|_{H^{k_q+1}(\Omega)} + \kappa^{-1/2} \ell_\Omega h^{k_q} |\mathbf{q}|_{H^{k_q+1}(\Omega)} + \kappa^{1/2} h^{k_u+1} |u|_{H^{k_u+1}(\Omega)}. \end{aligned}$$

It is observed that the error term in $\mathbf{q}$ and in u are of the same order if $k_q = k_u + 1$. In this case, this estimate is of order $\mathcal{O}(h^{k_q})$. However, this is in general pessimistic for $\|\nabla \cdot \mathbf{q} - \nabla \cdot \mathbf{q}_h\|$. For example,

for the Raviart-Thomas element $RT_0 \times P_0$ we have that $k_q = k_u = 0$, and thus there is no convergence rate for $\|\nabla \cdot \mathbf{q} - \nabla \cdot \mathbf{q}_h\|$. However, one can show that, in fact,

$$\|\nabla \cdot \mathbf{q} - \nabla \cdot \mathbf{q}_h\| \lesssim h |\nabla \cdot \mathbf{q}|_{H^1(\Omega)}.$$

In general, and with respect to the primal formulation, the dual formulation provides a better approximation to the dual variable $\mathbf{q}$ at the expense of lowering the convergence rate of the primal variable u . In some applications, such as the flow in porous media, this is important, as the variable that actually matters is $\mathbf{q}$.

1.5 Discontinuous Galerkin method for Poisson's problem

In this section we introduce the discontinuous Galerkin (dG) method. The idea is that the interpolating functions are *discontinuous* piecewise polynomials, and thus the resulting FE space is *non-conforming* in $H^1(\Omega)$, according to the definition of conformity given. In some cases, it is possible to use non-conforming approximations without modifying the bilinear form of the problem; this lack of conformity is one of Strang's variational crimes. However, most often discontinuous approximations require the modification of the bilinear form of the problem by introducing terms that involve jumps of the unknowns.

In these notes we will introduce a *non-standard* motivation to derive dG approximations, in particular for Poisson's problem in this chapter. The starting point will be a hybrid *conforming* approximation of the problem, from which the dG method will be derived for the irreducible problem.

1.5.1 An abstract elliptic problem in hybrid form

Hybrid continuous problem

The numerical approximation we propose can be motivated from a **hybrid formulation** of the problem. Broadly speaking, a hybrid formulation is a particular case of mixed formulation, in which there are unknowns defined only on the boundaries of the subdomains that conform a partition of Ω . In the FE context, these unknowns are defined on the edges of the FE partition.

Let us now introduce the hybrid formulation at the continuous level, prior to any numerical approximation. Assume that we have an abstract boundary value problem posed in a domain Ω , split as $\bar{\Omega} = \bar{\Omega}_1 \cup \bar{\Omega}_2$, and with $\Gamma = \partial\Omega_1 \cap \partial\Omega_2$ (see Fig. 1.1).

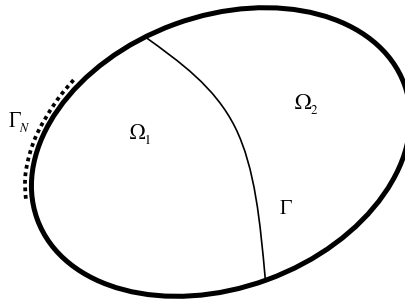


Figure 1.1: Splitting of Ω into two subdomains

Let X be a space of functions defined on Ω and consider an abstract variational problem consisting of finding an unknown u in X such that

$$a(u, v) = l(v) \quad \forall v \in X, \quad (1.98) \quad \{\text{eq:abs}\}$$

where $a(u, v)$ is a bilinear form on $X \times X$ and l a linear form defined on X , both assumed to be continuous. If the problem considered involves Dirichlet boundary conditions, we will assume that they are homogeneous and enforced in an essential manner in X .

Let u_i, v_i be the restrictions of $u, v \in X$ to subdomain Ω_i , and X_i the spaces where they belong, $i = 1, 2$. Suppose that $u \in X$ has a well defined trace on Γ belonging to a space T , and a flux corresponding to the differential operator associated to (1.98) belonging to a space F_i when computed from subdomain Ω_i , $i = 1, 2$. Then, the hybrid formulation of (1.98) that we consider is the following: find $u_i \in X_i$, $\lambda_i \in F_i$, $i = 1, 2$, and $\gamma \in T$ such that

$$\begin{aligned} a_1(u_1, v_1) - \langle \lambda_1, v_1 \rangle_\Gamma &= l_1(v_1) \quad \forall v_1 \in X_1, \\ a_2(u_2, v_2) - \langle \lambda_2, v_2 \rangle_\Gamma &= l_2(v_2) \quad \forall v_2 \in X_2, \\ \langle \mu_1, u_1 - \gamma \rangle_\Gamma &= 0 \quad \forall \mu_1 \in F_1, \\ \langle \mu_2, u_2 - \gamma \rangle_\Gamma &= 0 \quad \forall \mu_2 \in F_2, \\ \langle \zeta, \lambda_1 + \lambda_2 \rangle_\Gamma &= 0 \quad \forall \zeta \in T, \end{aligned}$$

where a_i and l_i are the restrictions of a and l to $X_i \times X_i$ and X_i , respectively.

If the problem includes imposition of fluxes of u in a part Γ_g of $\partial\Omega$, which for the sake of simplicity we may consider contained in $\partial\Omega_1$ (see Fig. 1.1, where $\Gamma_N \equiv \Gamma_g$), this imposition may be also “hybridised”, yielding the problem

$$\{\text{eq:hyb1}\} \quad a_1(u_1, v_1) - \langle \lambda_1, v_1 \rangle_\Gamma - \langle \lambda_g, v_1 \rangle_{\Gamma_g} = l_1(v_1) \quad \forall v_1 \in X_1, \quad (1.99)$$

$$\{\text{eq:hyb2}\} \quad a_2(u_2, v_2) - \langle \lambda_2, v_2 \rangle_\Gamma = l_2(v_2) \quad \forall v_2 \in X_2, \quad (1.100)$$

$$\{\text{eq:hyb3}\} \quad \langle \mu_1, u_1 - \gamma \rangle_\Gamma + \langle \mu_1, u_1 - \gamma \rangle_{\Gamma_g} = 0 \quad \forall \mu_1 \in F_1, \quad (1.101)$$

$$\{\text{eq:hyb4}\} \quad \langle \mu_2, u_2 - \gamma \rangle_\Gamma = 0 \quad \forall \mu_2 \in F_2, \quad (1.102)$$

$$\{\text{eq:hyb5}\} \quad \langle \zeta, \lambda_1 + \lambda_2 \rangle_\Gamma + \langle \zeta, \lambda_g \rangle_{\Gamma_g} = \langle \zeta, q \rangle_{\Gamma_g} \quad \forall \zeta \in T, \quad (1.103)$$

where q is the flux to be prescribed. In this case, the linear form l (and the forms l_1 and l_2 resulting from the splitting of the domain) does *not* include the prescription of the fluxes, and T includes functions defined on Γ_g .

Concerning the well posedness of the problem (for simplicity with $\Gamma_g = \emptyset$), let us write it as a single variational equation of the form

$$\{\text{eq:hybgen}\} \quad a_{\text{hyb}}(u_1, u_2, \lambda_1, \lambda_2, \gamma; v_1, v_2, \mu_1, \mu_2, \zeta) = l_{\text{hyb}}(v_1, v_2, \mu_1, \mu_2, \zeta), \quad (1.104)$$

where

$$\begin{aligned} a_{\text{hyb}}(u_1, u_2, \lambda_1, \lambda_2, \gamma; v_1, v_2, \mu_1, \mu_2, \zeta) &:= a_1(u_1, v_1) - \langle \lambda_1, v_1 \rangle_\Gamma + a_2(u_2, v_2) - \langle \lambda_2, v_2 \rangle_\Gamma \\ &\quad + \langle \mu_1, u_1 - \gamma \rangle_\Gamma + \langle \mu_2, u_2 - \gamma \rangle_\Gamma + \langle \zeta, \lambda_1 + \lambda_2 \rangle_\Gamma, \\ l_{\text{hyb}}(v_1, v_2, \mu_1, \mu_2, \zeta) &:= l_1(v_1) + l_2(v_2), \end{aligned}$$

with $(u_1, u_2, \lambda_1, \lambda_2, \gamma) \in X_1 \times X_2 \times F_1 \times F_2 \times T$ and which must hold for all $(v_1, v_2, \mu_1, \mu_2, \zeta) \in X_1 \times X_2 \times F_1 \times F_2 \times T$. If we assume that $a_i(u_i, v_i)$ are coercive in $X_i \times X_i$ ($i = 1, 2$), we have that

$$a_{\text{hyb}}(u_1, u_2, \lambda_1, \lambda_2, \gamma; u_1, u_2, \lambda_1, \lambda_2, \gamma) = a_1(u_1, u_1) + a_2(u_2, u_2) \gtrsim \|u_1\|_{X_1}^2 + \|u_2\|_{X_2}^2.$$

Control on $\|\lambda_1\|_{F_1}$, $\|\lambda_2\|_{F_2}$ and $\|\gamma\|_T$ can be obtained only if the inf-sup conditions

$$\inf_{\mu_i \in F_i \setminus \{0\}} \sup_{v_i \in V_i \setminus \{0\}} \frac{\langle \mu_i, v_i \rangle_\Gamma}{\|\mu_i\|_{F_i} \|v_i\|_{X_i}} \geq C_F, \quad i = 1, 2, \quad (1.105)$$

$$\inf_{\zeta \in T \setminus \{0\}} \sup_{\mu_i \in F_i, \mu_1 + \mu_2 \neq 0} \frac{\langle \zeta, \mu_1 + \mu_2 \rangle_\Gamma}{\|\zeta\|_T (\|\mu_1\|_{F_1} + \|\mu_2\|_{F_2})} \geq C_T, \quad (1.106)$$

hold for positive constants C_F and C_T . Under these assumptions (1.105) and (1.106), the coercivity of $a_i(u_i, v_i)$ in $X_i \times X_i$ ($i = 1, 2$) and appropriate continuity of all the forms involved in the problem, it can be proved using the same proof technique as in Theorem 10 that problem (1.104) satisfies an inf-sup condition that guarantees existence and uniqueness of solution by virtue of Theorem 8. When $a_i(u_i, v_i)$ are not coercive, but the problem corresponds to a well posed saddle point problem, this well posedness can also be established.

Finite element approximation

Recall that $\mathcal{E}_h = \{E\}$ denotes the collection of edges associated to $\mathcal{T}_h$. In particular, we will be interested in edges interior to Ω , which will be labelled E_0 , and edges on Γ_g , which will be labelled E_g . Summation over all the elements will be indicated as $\sum_K$ and summation over the edges as $\sum_E$.

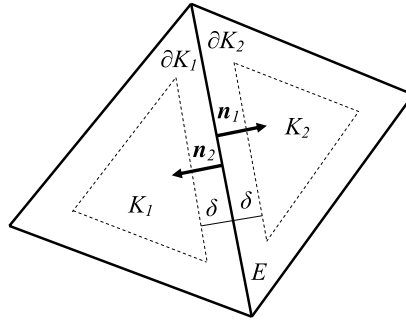


Figure 1.2: Notation for the approximation of fluxes and traces for interior edges

{fig:element}

Suppose now that elements K_1 and K_2 share an edge E , and let $\mathbf{n}_1$ and $\mathbf{n}_2$ be the normals to E exterior to K_1 and K_2 , respectively (see Fig. 1.2). For a scalar function f , possibly discontinuous across E , we define its jump as $\llbracket \mathbf{n} f \rrbracket_E := \mathbf{n}_1 f|_{\partial K_1 \cap E} + \mathbf{n}_2 f|_{\partial K_2 \cap E}$, and for a vector or tensor $\mathbf{v}$, $\llbracket \mathbf{n} \cdot \mathbf{v} \rrbracket_E := \mathbf{n}_1 \cdot \mathbf{v}|_{\partial K_1 \cap E} + \mathbf{n}_2 \cdot \mathbf{v}|_{\partial K_2 \cap E}$. Averages of functions discontinuous across an edge E will be denoted by $\{f\}_E = \frac{1}{2}(f|_{\partial K_1 \cap E} + f|_{\partial K_2 \cap E})$, omitting the subscript E when there is no possibility of confusion. Note that we could write $\llbracket \mathbf{n} f \rrbracket_E = 2\{\mathbf{n} f\}_E$, although we will keep the notation $\llbracket \mathbf{n} f \rrbracket_E$ to stress the presence of the normal $\mathbf{n}$, which gives the meaning of jump to this expression. Note also that it is common to omit the normal in the definition of $\llbracket \mathbf{n} f \rrbracket_E$, so that the jump of a scalar function is a vector.

We can easily extend formulation (1.99)-(1.103) to multiple subdomains and, in particular, to the case in which each of these is an element domain of $\mathcal{T}_h$. The resulting hybrid formulation can then be approximated using a FE method. Let V_h , F_h and T_h be the spaces where u_h , λ_h and γ_h are sought, respectively. Note that F_h is made of functions that may be multi-valuate on the edges of the FE partition, whereas T_h is made of functions uniquely defined on these edges. These spaces will be endowed with the appropriate norm depending on the problem considered.

The extension of the hybrid formulation to the elements of the FE partition reads as follows: find $(u_h, \lambda_h, \gamma_h) \in V_h \times F_h \times T_h$ such that

$$\sum_K a_K(u_h, v_h) - \sum_K \langle \lambda_h, v_h \rangle_{\partial K} = \sum_K l_K(v_h), \quad (1.107) \quad \{\text{eq:hfe}\}$$

$$\sum_K \langle \zeta_h, \lambda_h \rangle_{\partial K} = \langle \zeta_h, q \rangle_{\Gamma_g}, \quad (1.108) \quad \{\text{eq:hfe}\}$$

$$\sum_K \langle \mu_h, u_h - \gamma_h \rangle_{\partial K} = 0, \quad (1.109) \quad \{\text{eq:hfe}\}$$

for all $(v_h, \mu_h, \zeta_h) \in V_h \times F_h \times T_h$.

From the hybrid to the irreducible form

The objective of what follows can be summarised by saying that we intend to design dG methods based on the hybrid FE approximation (1.107)-(1.109). To this end, the key point to step to the method we propose is *to design expressions to compute the fluxes λ_h and the traces γ_h from finite difference-like approximations*. The steps we propose are:

1. Compute the fluxes λ_h from values of u_h, γ_h using finite difference-like expressions.
2. Compute the traces γ_h by imposing continuity of total fluxes, including Neumann boundary conditions where they are applied.
3. Go to the expression obtained in 1 with the expression of γ_h just obtained to compute λ_h in terms of u_h .

This general procedure will be applied in this chapter to Poisson's problem. Before proceeding, let us remark that step 2 will allow us to significantly simplify the problem. Suppose that $\lambda_h = \Lambda_h(u_h)$ is the expression of the fluxes in terms of u_h obtained, for simplicity without Neumann boundary conditions. If we can satisfy exactly the continuity of these fluxes *and take the flux test function* $\mu_h = \Lambda_h(v_h)$, v_h being the test function for u_h , then (1.108) will be exactly satisfied and $\sum_K \langle \mu_h, \gamma_h \rangle_{\partial K} = 0$. The problem reduces then to find $u_h \in V_h$ such that

$$\sum_K a_K(u_h, v_h) - \sum_K \langle \Lambda_h(u_h), v_h \rangle_{\partial K} - \sum_K \langle \Lambda_h(v_h), u_h \rangle_{\partial K} = \sum_K l_K(v_h), \quad (1.110) \quad \{\text{eq:hfef}\}$$

for all $v_h \in V_h$, where we have added up (1.107) and what remains of (1.109) because μ_h is not independent of v_h . The sign of the third term in the left-hand-side of this expression has been chosen to get a symmetric problem when B_K is symmetric.

A possible way to satisfy the exact continuity of fluxes is to take them of the form $\lambda_h = \Lambda_h(u_h) = \mathbf{n} \cdot \mathbf{\Lambda}_h^0(u_h)$, with $\mathbf{\Lambda}_h^0(u_h)$ a single-valued vector field defined on the element edges. This will be precisely the form of the fluxes we will propose.

When there are Neumann boundary conditions, a particular expression for the fluxes λ_h will be obtained, the corresponding test function being zero. This results in the classical boundary term corresponding to the prescription of fluxes that needs to be added to (1.110).

Remark 4 *Most dG methods are motivated by integration-by-parts of the differential equation tested with a test function v_h within each element, which leads to the first term in the left-hand-side of (1.110) and, as we shall see, part of the second. Then (part of) the third term is introduced to ensure symmetry, provided $\llbracket \mathbf{n}u \rrbracket_E = 0$ for the solution of the continuous problem u . After this, additional “stability”*

terms need to be added to ensure appropriate continuity of the unknown (in a weak sense). In our case, these terms will appear naturally in the definition of $\Lambda_h(u_h)$, whose design will be “non-classical” in order to satisfy (1.108) exactly. Moreover, even though it does not appear in (1.110), we will have at our disposal a definition of a trace.

Other hybrid formulations can be used as starting point to design dG methods. Our unknowns are the field in the interior of the elements and its traces and fluxes on the boundaries, but there are alternatives.

1.5.2 Application to Poisson's problem

{sec:cd}

Problem statement

The boundary value problem we consider is a slight modification of problem (1.1)-(1.3): find $u : \Omega \rightarrow \mathbb{R}$ such that

$$\begin{aligned} \mathcal{L}(u) &:= -\nabla \cdot (\kappa \nabla u) + \sigma u = f && \text{in } \Omega, \\ u &= 0 && \text{on } \Gamma_u, \\ \mathcal{F}(u) &:= \kappa \mathbf{n} \cdot \nabla u \equiv \kappa \partial_n u = q && \text{on } \Gamma_g, \end{aligned} \tag{1.111} \quad \{\text{eq:nbc}\}$$

The coefficients κ and σ may now be considered variable, although we will take them piecewise constant within each element of the FE partition, for simplicity. Note that the boundary condition (1.111) has been simplified with respect to (1.3) and we have introduced the flux operator $\mathcal{F}$.

The irreducible weak form of the problem is given by the straightforward modification of the variational problem (1.11).

The hybrid formulation corresponding to (1.99)-(1.103) in this case is posed in the spaces V defined as for the irreducible problem, $T = H_{00}^{1/2}(\Gamma)$ and $F_i = (H_{00}^{1/2}(\Gamma_i))'$ (the prime denoting dual space), with $\Gamma_i = \partial\Omega_i \cap \Omega$. The solution of the hybrid problem is $\gamma = u_1|_{\Gamma_1} = u_2|_{\Gamma_2}$, $\lambda_1 = -\lambda_2 = \kappa_1 \mathbf{n}_1 \cdot \nabla u_1|_{\Gamma_1} = -\kappa_2 \mathbf{n}_2 \cdot \nabla u_2|_{\Gamma_2}$. The extension to a hybrid problem considering all subdomains of the FE partition (problem (1.107)-(1.109)) is also immediately set. Abusing of the notation, we can consider this same problem as a *FE approximation* of the hybrid continuous one, the FE spaces being

$$\begin{aligned} V_h &= \left\{ v_h : \bigcup K \rightarrow \mathbb{R} \mid v_h|_K \in P_{k_u}(K), K \in \mathcal{T}_h \right\}, \\ F_h &= \left\{ \mu_h : \bigcup \partial K \rightarrow \mathbb{R} \mid \mu_h|_{\partial K} \in P_{k_f}(\partial K), K \in \mathcal{T}_h \right\}, \\ T_h &= \left\{ \zeta_h : \bigcup E \rightarrow \mathbb{R} \mid \zeta_h|_E \in P_{k_t}(E), E \in \mathcal{E}_h, \text{ and } \zeta_h|_E = 0 \text{ if } E \subset \Gamma_u \right\}, \end{aligned}$$

where k_u , k_f and k_t are respectively the interpolation orders for the unknown u , the fluxes and the traces. These spaces have been constructed directly based on the subdomains of the FE partition, not in the reference domains. We assume that the mapping φ_K introduced earlier in Eq. (1.23) is affine.

Approximation of fluxes and calculation of traces

Assume first that we have an interior edge, the situation depicted in Fig. 1.2. Suppose that this edge E is shared by two elements $K_1, K_2 \in \mathcal{T}_h$. Let us take now a normal direction across edge E , either $\mathbf{n}_1$ or $\mathbf{n}_2$. Values of the unknowns and the coefficient κ will be denoted with the superscript $-$ or $+$ depending on whether they belong to the left or right side of E with respect to the normal chosen.

We assume that the approximation u_h is meaningful to compute the fluxes λ_h up to a distance δ to edge E , on which the trace takes a value γ_h . The distance δ will be a parameter of the formulation (see

Fig. 1.3 for a sketch in a one-dimensional case with linear interpolation). Using an approximation of order $\mathcal{O}(\delta)$, the fluxes can be approximated as

$$\lambda_h^\pm \approx \frac{\kappa^\pm}{\delta}(\gamma_h - u_{h,\delta}^\pm),$$

where $u_{h,\delta}^+$ and $u_{h,\delta}^-$ are the values of u_h at a distance δ from E . These can be computed from a Taylor expansion as (see Figure 1.3):

$$u_{h,\delta}^\pm = u_h^\pm - \delta \partial_{n^\pm} u_h^\pm + \mathcal{O}(\delta^2),$$

where ∂_{n^+} is the derivative in the direction of the normal chosen and ∂_{n^-} the opposite. Therefore:

$$\lambda_h^\pm \approx \frac{\kappa^\pm}{\delta}(\gamma_h - u_h^\pm) + \kappa^\pm \partial_{n^\pm} u_h^\pm. \quad (1.112)$$

Equation (1.112) completes the first step of the strategy described earlier. It is obvious that other possibilities could be used. In particular, if $u_{h,i}$ is the value of u_h at a distance $i\delta$ from the edge, $i = 1, 2, \dots$, we could approximate

$$\lambda_h^\pm = \frac{\kappa^\pm}{\delta} \left(w_0 \gamma_h - \sum_{i=1}^q w_i u_{h,i}^\pm \right) + \mathcal{O}(\delta^q),$$

where $q \geq 1$ and the weights w_i , $i = 0, 1, \dots, q$, satisfy the consistency conditions $\sum_{i=0}^q w_i = 0$, $\sum_{i=0}^q i w_i = 1$. Using the expansions within an element

$$u_{h,i}^\pm = u_h^\pm + \sum_{j=1}^p \frac{1}{j!} (-i\delta)^j \partial_{n^\pm}^j u_h^\pm + \mathcal{O}(\delta^{p+1}),$$

with $p \geq q$, we get

$$\lambda_h^\pm = \frac{\kappa^\pm w_0}{\delta} (\gamma_h - u_h^\pm) + \kappa^\pm \partial_{n^\pm} u_h^\pm + \mathcal{O}(\delta^q).$$

The change with respect to (1.112) is only in the factor that multiplies $1/\delta$, which is a numerical parameter. Thus, we may keep (1.112) for our purposes, noting that *the finite difference approximation does not affect the resulting approximation*, provided only degrees of freedom within one element are used. Of course, other methods could be devised if $u_{h,\delta}^+$ and $u_{h,\delta}^-$ are approximated using neighbouring elements. The final stencil in this case would be wider.

The second step consists of imposing continuity of *total* fluxes. For the continuous problem, this amounts to say that $\llbracket \mathbf{n} \cdot \kappa \nabla u \rrbracket \equiv \llbracket \mathcal{F}(u) \rrbracket = 0$. Note that here we use the symbol $\llbracket \cdot \rrbracket$ understanding that $\mathcal{F}(u)$ contains the normal in its definition. Using expressions (1.112) and going back to the notation in Fig. 1.2, the approximate continuity of fluxes leads to

$$\begin{aligned} 0 &= \llbracket \mathcal{F}(u) \rrbracket \\ &\approx \lambda_{h,1} + \lambda_{h,2} \\ &\approx \frac{1}{\delta} [(\kappa_1 + \kappa_2) \gamma_h - \kappa_1 u_{h,1} - \kappa_2 u_{h,2}] + \llbracket \mathcal{F}(u_h) \rrbracket \\ &= \llbracket \mathcal{F}(u_h) \rrbracket + \frac{2}{\delta} (\{\kappa\} \gamma_h - \{\kappa u_h\}). \end{aligned} \quad (1.113)$$

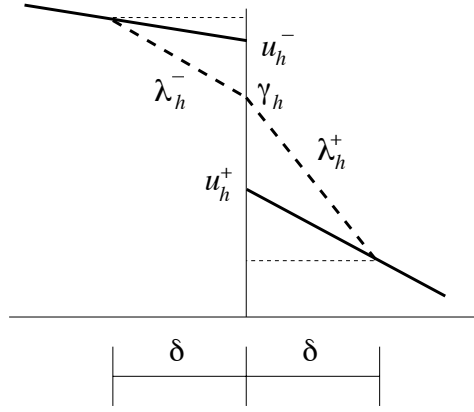


Figure 1.3: Approximation of fluxes in a one-dimensional case with linear interpolation. $u_h^\pm$ are the values of u_h from both sides of the element boundary (now a point), γ_h is the trace and $\lambda_h^\pm$ are the fluxes reconstructed from the trace and the values of u_h and a distance δ from the boundary, computed as the slope of the thick dotted line multiplied by the appropriate coefficient κ

This allows us to obtain the approximation of the traces on each edge we were looking for:

$$\gamma_h = \frac{\{\kappa u_h\}}{\{\kappa\}} - \frac{\delta}{2\{\kappa\}} \llbracket \mathcal{F}(u_h) \rrbracket. \quad (1.114)$$

Here and below, we will use the symbol $=$ for our models to compute fluxes and traces, understanding that they are approximations to the exact expressions.

Inserting (1.114) into (1.112) we see that the fluxes on the boundaries of the elements can be computed as

$$\begin{aligned} \lambda_h &= \frac{\kappa}{\delta} (\gamma_h - u_h) + \mathcal{F}(u_h) \\ &= \frac{\kappa}{\delta} \left(\frac{\{\kappa u_h\}}{\{\kappa\}} - u_h \right) - \frac{\kappa}{2\{\kappa\}} \llbracket \mathcal{F}(u_h) \rrbracket + \mathcal{F}(u_h). \end{aligned}$$

This expression can be written in a more suggestive way. Consider the harmonic average:

$$\langle \kappa \rangle := \frac{\kappa_1 \kappa_2}{\{\kappa\}} = \frac{2\kappa_1 \kappa_2}{\kappa_1 + \kappa_2}.$$

We may write, after some algebraic manipulations (considering for example element K_1):

$$\begin{aligned} \frac{\kappa_1}{\delta} \left(\frac{\{\kappa u_h\}}{\{\kappa\}} - u_{h,1} \right) &= -\frac{\langle \kappa \rangle}{2\delta} \mathbf{n}_1 \cdot \llbracket \mathbf{n} u_h \rrbracket, \\ -\frac{\kappa_1}{2\{\kappa\}} \llbracket \mathcal{F}(u_h) \rrbracket + \mathcal{F}(u_h)_1 &= \langle \kappa \rangle \mathbf{n}_1 \cdot \{\nabla u_h\}. \end{aligned}$$

We finally obtain:

$$\lambda_h = -\frac{\langle \kappa \rangle}{2\delta} \mathbf{n} \cdot \llbracket \mathbf{n} u_h \rrbracket + \langle \kappa \rangle \mathbf{n} \cdot \{\nabla u_h\}. \quad (1.115) \quad \{\text{eq:fluxes}\}$$

Equations (1.114) and (1.115) are the expressions sought in the case of interior edges. Note that (1.115) is in fact defined on the boundaries of the elements, and therefore the value on the edge is not uniquely

defined. In fact, as it was mentioned earlier, these fluxes are of the form $\lambda_h = \mathbf{n} \cdot \mathbf{\Lambda}_h^0(u_h)$, with $\mathbf{\Lambda}_h^0(u_h) = -\frac{\langle \kappa \rangle}{2\delta} \llbracket \mathbf{n} u_h \rrbracket + \langle \kappa \rangle \{ \nabla u_h \}$ single-valued.

Suppose now that the edge we consider is on the Neumann boundary, that is to say, $E \subset \Gamma_g$. The continuity of total fluxes has to be replaced by the *equilibrium* of fluxes in the case of Neumann edges. Proceeding as above, we now have that

$$\lambda_h = q. \quad (1.116) \quad \{\text{eq:flu}\}$$

This expression is defined for the part of the element boundaries that belongs to Γ_g . From the approximation of λ_h , which now does not depend on the traces, we have:

$$q = \lambda_h \approx \mathcal{F}(u_h) + \frac{\kappa}{\delta}(\gamma_h - u_h).$$

Therefore:

$$\gamma_h = u_h - \frac{\delta}{\kappa}(\mathcal{F}(u_h) - q). \quad (1.117) \quad \{\text{eq:tracesN}\}$$

The resulting discontinuous Galerkin approximation

The original hybrid formulation can be written in a compact form as indicated in (1.110) with $\Lambda(v_h)$ given by (1.115) or (1.116).

Let us collect the expressions obtained. The fluxes and traces for interior edges are given by:

$$\gamma_h = \frac{\{\kappa u_h\}}{\{\kappa\}} - \frac{\delta}{2\{\kappa\}}(\llbracket \kappa \mathbf{n} \cdot \nabla u_h \rrbracket), \quad (1.118) \quad \{\text{eq:ut}\}$$

$$\lambda_h = -\frac{\langle \kappa \rangle}{2\delta} \mathbf{n} \cdot \llbracket \mathbf{n} u_h \rrbracket + \langle \kappa \rangle \mathbf{n} \cdot \{ \nabla u_h \}. \quad (1.119) \quad \{\text{eq:uf}\}$$

The test functions may be taken as

$$\zeta_h = \frac{\{\kappa v_h\}}{\{\kappa\}} - \frac{\delta}{2\{\kappa\}} \llbracket \kappa \mathbf{n} \cdot \nabla v_h \rrbracket, \quad (1.120) \quad \{\text{eq:tt}\}$$

$$\mu_h = -\frac{\langle \kappa \rangle}{2\delta} \mathbf{n} \cdot \llbracket \mathbf{n} v_h \rrbracket + \langle \kappa \rangle \mathbf{n} \cdot \{ \nabla v_h \}, \quad (1.121) \quad \{\text{eq:tf}\}$$

with $v_h \in V_h$.

For Neumann edges the corresponding expressions are:

$$\begin{aligned} \gamma_h &= u_h - \frac{\delta}{\kappa}(\kappa \mathbf{n} \cdot \nabla u_h - q), \\ \lambda_h &= q, \\ \zeta_h &= v_h - \delta \mathbf{n} \cdot \nabla v_h, \\ \mu_h &= 0. \end{aligned}$$

Remark 5 *It is obvious that other choices of test functions are possible, particularly if no care is put on whether the resulting formulation is symmetric or not. The ones chosen here automatically will yield adjoint consistency of the method to be designed.*

We have that

$$- \sum_K \langle \lambda_h, v_h \rangle_{\partial K} + \sum_K \langle \zeta_h, \lambda_h \rangle_{\partial K} - \sum_K \langle \mu_h, u_h - \gamma_h \rangle_{\partial K}$$

$$\begin{aligned}
&= - \sum_E \left\langle -\frac{\langle \kappa \rangle}{2\delta} \llbracket \mathbf{n} u_h \rrbracket + \langle \kappa \rangle \{ \nabla u_h \}, \llbracket \mathbf{n} v_h \rrbracket \right\rangle_E \\
&\quad - \sum_E \left\langle -\frac{\langle \kappa \rangle}{2\delta} \llbracket \mathbf{n} v_h \rrbracket + \langle \kappa \rangle \{ \nabla v_h \}, \llbracket \mathbf{n} u_h \rrbracket \right\rangle_E \\
&\quad - \langle v_h, q \rangle_{\Gamma_g} + \langle \zeta_h, q \rangle_{\Gamma_g} \\
&= \sum_E \langle \kappa \rangle \delta \left\langle \frac{1}{\delta} \llbracket \mathbf{n} v_h \rrbracket - \{ \nabla v_h \}, \frac{1}{\delta} \llbracket \mathbf{n} u_h \rrbracket - \{ \nabla u_h \} \right\rangle_E - \sum_E \langle \kappa \rangle \delta \langle \{ \nabla v_h \}, \{ \nabla u_h \} \rangle_E \\
&\quad - \langle v_h, q \rangle_{\Gamma_g} + \langle \zeta_h, q \rangle_{\Gamma_g}
\end{aligned}$$

Using this in (1.110) we obtain the method we propose:

Find $u_h \in V_h$ such that

$$\begin{aligned}
&\sum_K \kappa \langle \nabla v_h, \nabla u_h \rangle_K + \sum_K \sigma \langle v_h, u_h \rangle_K \\
&\quad + \sum_E \langle \kappa \rangle \delta \left\langle \frac{1}{\delta} \llbracket \mathbf{n} v_h \rrbracket - \{ \nabla v_h \}, \frac{1}{\delta} \llbracket \mathbf{n} u_h \rrbracket - \{ \nabla u_h \} \right\rangle_E - \sum_E \langle \kappa \rangle \delta \langle \{ \nabla v_h \}, \{ \nabla u_h \} \rangle_E \\
&= \sum_K l_K(v_h) + \langle v_h, q \rangle_{\Gamma_g}, \tag{1.122}
\end{aligned}$$

for all $v_h \in V_h$.

Remark 6 Let us make several remarks about the numerical approximation just obtained:

- Essentially, we have recovered the so called **interior penalty method**. Note that we may take $\delta = \delta_0 h$, and now δ_0 can be taken as the algorithmic parameter of the formulation. It has to be small enough, so that the region of each element left when the layer of width δ is subtracted is not empty (see Fig. 1.2). This penalty term is not added “a posteriori”, as usual, but comes from the reconstruction of the unknown in terms of its values in the element interiors and its traces on the element boundaries. We have not analyzed the conditions on δ_0 under which the method is optimal.
- Adjoint consistency is ensured. This is a result of taking the test functions as indicated in (1.120) and (1.121), and therefore it is quite natural when looking at the hybrid form of the problem.
- Several different methods may be devised within the framework presented. Among the possible alternatives there are different approximations of the diffusive fluxes using finite-difference-like expressions or other expressions of the test functions. Note however that if in (1.107)-(1.109) we do not take the test functions as $\mu_h = \Lambda(v_h)$, more terms than in (1.110) will appear.
- The formulation obtained is intrinsically consistent. Therefore, the order of accuracy will depend on the interpolation space V_h chosen for the primal variable.
- The most salient feature of the method obtained with respect to the classical interior penalty method is the use of harmonic averages rather than arithmetic ones. This has been derived from the continuity of the fluxes.

Regardless of the derivation employed, problem (1.122) is the most common dG approximation found in the literature. Apart from the possibility of using it as a general approximation scheme, it can

be applied also in cases where conformity is difficult to employ, as in the case of **hanging nodes**. These are nodes of one element that belong to the edge of a neighbour, and appear for example when one element is subdivided in a refinement process while the neighbour is not.